

鑫拓智能 | EXPANDING INTELLIGENCE

SITO Strain Wave Drive Reducers Smart Actuator

STRAIN WAVE DRIVE REDUCERS

Product Selection Catalog

Strain Wave Gear Catalog

www.sitodrive.com

STRAIN WAVE DRIVE REDUCERS

Amid the wave of intelligent manufacturing and the development of new quality productive forces, Guangdong SITO Drive Technology Co., Ltd., located in the prime inner bay of the Greater Bay Area, has rapidly emerged as a leading supplier of harmonic reducers and core power unit servo joint modules in the field of intelligent manufacturing. This growth is driven by years of advanced technology in military-grade precision component manufacturing, supported by targeted research from national laboratories in basic materials and harmonic gear transmission, and fueled by the company's business philosophy: "Innovation Leads the Future, Quality Shapes the Lifeline, and Specialization Continues the Legacy."

SITO Harmonic Drive is committed to ultra-high positioning accuracy and long-lasting durability as its core technical beliefs, and is dedicated to delivering power unit performance that exceeds customer expectations.

With micron-level transmission precision, we power the new era of intelligent connectivity.

Focusing on niche markets, SITO is deeply engaged in the R&D and application of compact, high-torque harmonic reducers.

Our strategic pillars include:

Standardized modular interfaces at the hardware level

Open development environments at the software level

Comprehensive online diagnostic systems with standardized teaching at the service level

A global sales and service network layout

We aspire to become a hidden champion in the field of precision transmission and to build a new ecosystem of human-machine collaboration.

All members of SITO uphold the core values of: "Excellence · Co-Creation · Vision."

Strain Wave Drive Reducers

CATEGORY **PREVIEW**

HD Series

PAGE: **11-28**

Model	Gear Ratio	PAGE
HD mini-03	30/50	17
HD mini-05	30/50/80	18
HD mini-08	30/50/100	19
HD mini-11	30/50/100	20
HD-14	30/50/80/100	21
HD-17	30/50/80/100	22
HD-20	30/50/80/100/120	23
HD-25	30/50/80/100/120	24
HD-11 Pro	50/100	25
HD-14 Pro	50/100	26
HD-17 Pro	50/100	27

HS Series

PAGE: **29-54**

Model	Gear Ratio	PAGE
HS-14	30/50/80/100	35
HS-17	30/50/80/100	39
HS-20	30/50/80/100/120	43
HS-25	30/50/80/100/120	47
HS-32	50/80/100/120	51

I Simplicity unit
II Hollow shaft
III Standard
IV Input shaft

CS Series

PAGE: **55-78**

Model	Gear Ratio	PAGE
CS-14	30/50/80/100	59
CS-17	30/50/80/100	61
CS-20	30/50/80/100/120	63
CS-25	30/50/80/100/120	65
CS-32	50/80/100/120	67

I Simplicity unit II Standard

CS mini-2.5	30/50	69
CS mini-03	30/50	72
CS mini-05	30/50/80	73
CS mini-08	30/50/100	75
CS mini-11	30/50/100	77

CD Series

PAGE: **79-85**

Model	Gear Ratio	PAGE
CD-14	**14:** 30/50/80/100	83
CD-17	**17:** 50/80/100	84
CD-20	**20:** 30/50/80/100/120	85

CORE **TECHNOLOGY**

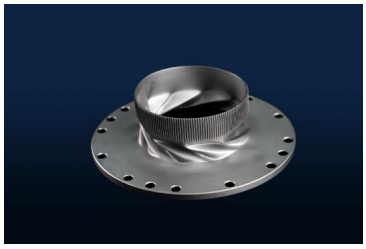
Strain Wave Drive Reducers



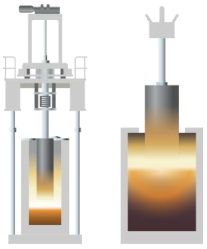
THE TOUGHNESS OF THE FLEX WHEEL

Resilience-Based Protection

Through specialized techniques such as electroslag remelting purification of raw materials and repeated precision spinning, we have significantly enhanced the stability of various performance indicators of the flexspline material.
This ensures that the stringent fatigue strength requirements of the flexspline are fully met. Under overload conditions, the material's toughness allows for plastic deformation, thereby providing maximum protection to the overall mechanical structure from damage.



Flexspline Resilience Protection Test



Electroslag Remelting Purification

SPECIAL MATERIALS BASIC R&D

Ultra-Fine Austempered Ductile Iron (ADI) Material

We are committed to fundamental materials research and have overcome multiple technical challenges.By applying isothermal quenching to ultra-. ne ductile iron hollow continuous cast profiles, we have successfully developed ultra-fine Austempered Ductile Iron (ADI) materials.

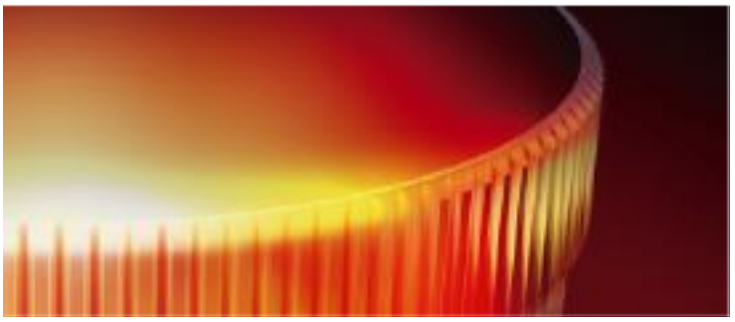
Key Features:1.Self-lubricating 2.Low temperature rise 3.Excellent vibration damping and noise reduction 4.Wide se ice temperature range 5.Extended se ice life



HEAT TREATMENT TECHNOLOGY

High-Precision Heat Treatment Technology

Over years of technical accumulation in heat treatment processes, we have identified the correlation and balance point between hardness and toughness—two critical factors affecting the fatigue strength of harmonic components.
By analyzing stress concentration after machining and stress dissipation before machining, we have developed a refined and comprehensive heat treatment process, which has now been fully implemented in mass production.

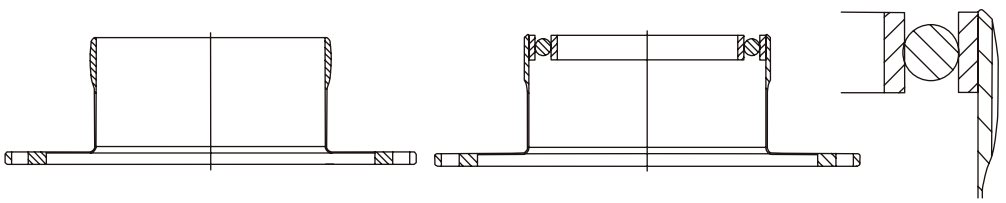


Material Heat Treatment Simulation

SPECIAL PROCESS OF FLEXIBLE PULLEY

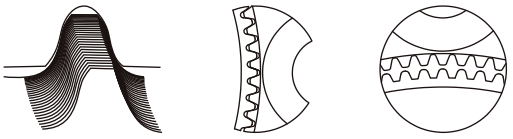
Special Processing Techniques for the Flexspline

We have innovatively developed a post-deformation finishing method in the flexspline tooth profile machining process.
This technique resolves the long-standing issue of uneven contact in traditional steel flexsplines while maintaining high precision.
As a result, the tooth surfaces achieve full meshing contact, with more uniform load distribution, reduced friction, and significantly enhanced durability of the product.



INNOVATIVE S-TOOTH DESIGN

In collaboration with university-based Deep-Sea Silent Transmission Laboratories, we conducted in-depth research on smoothing friction transmission and reducing the friction coefficient during gear cutting.
As a result, we innovatively developed the S-shaped harmonic tooth profile design theory, integrating principles from kinematics, tribology, and lubrication science.
Additionally, we designed and implemented a simulation software capable of automatically generating S-shaped tooth curves.



MACHINING TOOL DESIGN

We have creatively developed an integrated compound cutting tool that combines pre-cutting and post-forming processes for harmonic gear machining.
This breakthrough solves the long-standing challenge of tool manufacturing for micro-sized harmonic reducers, effectively addressing a gap in the market.
As a result, we have successfully launched a series of proprietary micro harmonic reducer products, pioneering a solution unique to the industry.



TESTING LABORATORY

In collaboration with leading universities and the government quality inspection bureau, we have established China's first comprehensive performance testing platform for harmonic reducers.
This platform enables full-spectrum testing, including service life, precision, starting torque, torsional stiffness, and load capacity, ensuring strict quality control throughout the product development process.
It provides robust experimental data to support each stage of R&D and safeguards the performance and reliability of harmonic drive products.



INSTALLATION AND PRECAUTIONS

We have innovatively developed a post-deformation finishing process in flexspline tooth profile machining. This breakthrough addresses the issue of uneven contact commonly found in traditional steel flexsplines, while maintaining high precision.As a result, the tooth surfaces achieve full and accurate meshing, leading to more uniform load distribution, reduced friction, and significantly enhanced product durability.

- Inject lubricant into the inner wall cavity of the flexspline, then install the reducer onto the output end and secure it with screws and spring washers for pre-tightening.

- Evenly apply lubricant to the flexible bearing, then install the wave generator into the flexspline and the output-end bearing.
  When installing the wave generator, special care should be taken to minimize the effects of center offset and misalignment, in order to reduce the impact of eccentricity.
  Avoid applying excessive force to the wave generator bearing area. The wave generator can be smoothly installed into the flexspline by rotating it gently during installation.
  After installation, gently rotate the wave generator by hand. If it rotates with consistent force, it can be considered normal.
  If there is significant unevenness, it indicates an asymmetric meshing state. In this case, the wave generator needs to be removed and reinstalled.

- Keep the motor rotating and tighten the screws on the output end in a crisscross pattern, gradually increasing the force in four to five equal increments until the desired tightening torque is reached.

Impo an t Notes

1. If the output end of the reducer is consistently positioned facing downward, the lubricant injected into the inner wall cavity of the exspline must extend beyond the meshing tooth surface.
2. A static seal must be applied between the installation planes of the rigid wheel and rigid wheel connectors, as well as between the installation planes of the flexspline and flexspline connectors. This is to ensure that grease does not leak during the use of the reducer, preventing damage to the reducer when operating with insufficient or no lubrication.
3. During installation, do not strike the reducer components with hard objects or apply excessive force, to avoid damage to the parts.
4. If you encounter any issues during the installation process, please contact us.

Full Series Technical Data Table for Strain Wave Gear

Series	Model	Ratio	Rate Torque at 2000 rpm [N.m]	Limit for Repeated Peak Torque [N.m]	Limit for Average Torque [N.m]	Limit for Momentary Peak Torque [N.m]	Maximum Input Speed [r/min]	Limit for Average Input Speed [r/min]	Moment of Inertia ($1/4GD^2$) [$kgcm^2$]	Backlash [Arc sec]	Life [hour]
HS	HS-14	30	4	9	6.8	17	8500	3500	0.033	≤20	7000
		50	5.4	18	6.9	35					
		80	7.8	23	11	47					10000
		100	7.8	28	11	54					
	HS-17	30	8.8	16	12	30	7300	3500	0.079	≤20	7000
		50	16	34	26	70					
		80	22	43	27	87				≤10	10000
		100	24	54	39	110					
	HS-20	30	15	27	20	50	6500	3500	0.193	≤20	7000
		50	25	56	34	98					
		80	34	74	47	127				≤10	10000
		100	40	82	49	147					
		120	40	87	49	147					
	HS-25	30	27	50	38	95	5600	3500	0.413	≤20	7000
		50	39	98	55	186					
		80	63	137	87	255				≤10	10000
		100	67	157	108	284					
		120	67	167	108	304					
	HS-32	50	76	216	108	382	4800	3500	1.69	≤20	7000
		80	118	304	167	568					10000
		100	137	333	216	647				≤10	
		120	137	353	216	686					
HD	HD mini-03	30	0.08	0.15	0.12	0.25	10000	6500	/	≤20	7000
		50	0.13	0.24	0.16	0.5					
	HD mini-05	30	0.23	0.47	0.35	0.85	10000	6500	/	≤20	7000
		50	0.38	0.87	0.5	1.7					
		80	0.5	1.2	0.82	2.2					10000
	HD mini-08	30	0.8	1.6	1.3	3.2	8500	3500	/	≤20	7000
		50	1.6	3.1	2.1	6.1					
		100	2.4	4.6	3.2	8.7					10000
	HD mini-11	30	2.1	4.1	3.1	8.2	8500	3500	/	≤20	7000
		50	3.2	7.8	5.1	15					
		100	4.8	10.5	8.7	23					10000
	HD-14	30	3	8.7	6.2	15	8500	3500	0.021	≤20	7000
		50	4.5	13	6.8	25					
		80	6	15	9.5	38				≤10	10000
		100	7.4	19	10.8	40					
	HD-17	30	6.6	13.8	10.8	28.8	7000	3500	0.054	≤20	7000
		50	11	23	18	48					
		80	12.8	29.6	21.6	56.8				≤10	10000
		100	16	37	27.8	71					
	HD-20	30	10.2	23.4	14.4	41.4	7000	3500	0.09	≤20	7000
		50	17	39	24	69					
		80	22.4	45.6	27.2	76				≤10	10000
		100	28	57	32	95					
		120	28	61	35	95					
	HD-25	30	16.2	41.4	22.8	76.2	5600	3500	0.282	≤20	7000
		50	26	69	38	127					
		80	44	95	60	147.2				≤10	10000
		100	48	110	75	184					
		120	54	120	80	202					

Series	Model	Ratio	Rate Torque at 2000 rpm [N.m]	Limit for Repeated Peak Torque [N.m]	Limit for Average Torque [N.m]	Limit for Momentary Peak Torque [N.m]	Maximum Input Speed [r/min]	Limit for Average Input Speed	Moment of Inertia ($1/4GD^2$) [$kgcm^2$]	Backlash [Arc sec]	Life [hour]
HD	HD-11 Pro	50	4.5	13	6.8	25	8500	3500	0.021	≤20	7000
		100	7.4	19	10.8	40				≤10	10000
	HD-14 Pro	50	11	23	18	48	7000	3500	0.054	≤20	7000
		100	16	37	27.8	71				≤10	10000
	HD-17 Pro	50	17	39	24	69	7000	3500	0.09	≤20	7000
		100	28	57	32					≤10	10000
CS	CS-14	30	4	9	6.8	17	8500	3500	0.033	≤20	7000
		50	5.4	18	6.9	35					
		80	7.8	23	9.5	47				≤10	10000
		100	7.8	28	11	54					
	CS-17	30	8.8	16	12	30	7300	3500	0.079	≤20	7000
		50	16	34	26	70					
		80	22	43	27	87				≤10	10000
		100	24	54	39	108					
	CS-20	30	15	27	20	50	6500	3500	0.193	≤20	7000
		50	25	56	34	98					
		80	34	74	47	121				≤10	10000
		100	40	82	49	147					
		120	47	87	55	160					
	CS-25	30	27	50	38	95	5600	3500	0.413	≤20	7000
		50	39	98	55	186					
		80	63	137	87	255				≤10	10000
		100	67	157	108	289					
		120	78	167	115	304					
	CS-32	50	76	216	108	382	4800	3500	1.69	≤20	7000
		80	118	304	167	568				≤10	10000
		100	137	353	216	647					
		120	152	353	240	686					
	CS Mini-2.5	30	0.06	0.13	0.1	0.21	7000	3500	/	≤20	7000
		50	0.10	0.21	0.13	0.40					
CS Mini	CS Mini-03	30	0.06	0.13	0.1	0.22	10000	6500	/	≤20	7000
		50	0.11	0.21	0.13	0.41					
	CS Mini-05	30	0.25	0.5	0.38	0.9	10000	6500	2.5x10^-4	≤20	7000
		50	0.4	0.9	0.53	1.8					
		80	0.52	1.35	0.84	2.5					10000
	CS Mini-08	30	0.9	1.8	1.4	3.3	8500	3500	3.2x10^-3	≤20	7000
		50	1.8	3.3	2.3	6.6					
		100	2.4	4.8	3.3	9					10000
	CS Mini-11	30	2.2	4.5	3.4	8.5	8500	3500	1.4x10^-2	≤20	7000
		50	3.5	8.3	5.5	17					
		100	5	11	8.9	25					10000
CD	CD-14	30	2.2	10.2	2.88	14.4	8500	3500	0.021	≤20	7000
		50	3.7	12	4.8	24					
		80	5.4	16	7.7	31				≤10	10000
		100	5.4	19	7.7	31					
	CD-17	50	11	23	18	48	7300	3500	0.054	≤20	7000
		80	15	29	19	55				≤10	10000
		100	16	37	27	55					
	CD-20	30	10.2	23.4	14.4	41.4	6500	3500	0.09	≤20	7000
		50	17	39	24	69					
		80	24	51	33	76				≤10	10000
		100	28	57	34	76					
		120	28	61	34	95					

HD

STRAIN WAVE DRIVE REDUCERS

HD Series

Model	Ratio	Page
HD mini-03	30/50	17
HD mini-05	30/50/80	18
HD mini-08	30/50/100	19
HD mini-11	30/50/100	20
HD-14	30/50/80/100	21
HD-17	30/50/80/100	22
HD-20	30/50/80/100/120	23
HD-25	30/50/80/100/120	24
HD-11 Pro	50/100	25
HD-14 Pro	50/100	26
HD-17 Pro	50/100	27



Features and Applications

The HD series is designed to achieve extreme flatness, with an axial length reduced by approximately 50% compared to the HS series. It is a simplified modular type featuring a high-rigidity cross roller bearing on the output side, making it ideal for applications that require a flat design.
At the same time, the development of the HD micro harmonic has expanded its applications in the robotics field.

- Ultra-thin
- Hollow Sha t
- Compact
- High Torque Capacity
- High Rigidity
- Coaxial Input and Output
- Excellent Positioning Accuracy and Rotational Precision

Product Model Description

HD-20-100

Model Series	Model	Ratio					-	-
HD	mini 03	30	50				-	-
	mini 05	30	50	80				
	mini 08	30	50		100			
	mini 11	30	50		100			
	14	30	50	80	100			
	17	30	50	80	100			
	20	30	50	80	100	120		
	25	30	50	80	100	120		
	11 pro		50		100			
	14 pro		50		100			
	17 pro		50		100			

Series	Model	Ratio	Rate Torque at 2000 rpm [N.m]	Limit for Repeated Peak Torque [N.m]	Limit for Average Torque [N.m]	Limit for Momentary Peak Torque [N.m]	Maximum Input Speed [r/min]	Limit for Average Input Speed [r/min]	Moment of Inertia ($1/4GD^2$) [$kgcm^2$]	Backlash [Arc sec]	Life [hour]
HD	HD mini-03	30	0.08	0.15	0.12	0.25	10000	6500	/	≤20	7000
		50	0.13	0.24	0.16	0.5					
	HD mini-05	30	0.23	0.47	0.35	0.85	10000	6500	/	≤20	7000
		50	0.38	0.87	0.5	1.7					
		80	0.5	1.2	0.82	2.2					10000
	HD mini-08	30	0.8	1.6	1.3	3.2	8500	3500	/	≤20	7000
		50	1.6	3.1	2.1	6.1					
		100	2.4	4.6	3.2	8.7					10000
	HD mini-11	30	2.1	4.1	3.1	8.2	8500	3500	/	≤20	7000
		50	3.2	7.8	5.1	15					
		100	4.8	10.5	8.7	23					10000
	HD-14	30	3	8.7	6.2	15	8500	3500	0.021	≤20	7000
		50	4.5	13	6.8	25					
		80	6	15	9.5	38				≤10	10000
		100	7.4	19	10.8	40					
	HD-17	30	6.6	13.8	10.8	28.8	7000	3500	0.054	≤20	7000
		50	11	23	18	48					
		80	12.8	29.6	21.6	56.8				≤10	10000
		100	16	37	27.8	71					
	HD-20	30	10.2	23.4	14.4	41.4	7000	3500	0.09	≤20	7000
		50	17	39	24	69					
		80	22.4	45.6	27.2	76				≤10	10000
		100	28	57	32	95					
		120	28	61	35	95					
	HD-25	30	16.2	41.4	22.8	76.2	5600	3500	0.282	≤20	7000
		50	26	69	38	127					
		80	44	95	60	147.2				≤10	10000
		100	48	110	75	184					
		120	54	120	80	202					
	HD-11 Pro	50	4.5	13	6.8	25	8500	3500	0.021	≤20	7000
		100	7.4	19	10.8	40				≤10	10000
	HD-14 Pro	50	11	23	18	48	7000	3500	0.054	≤20	7000
		100	16	37	27.8	71				≤10	10000
	HD-17 Pro	50	17	39	24	69	7000	3500	0.09	≤20	7000
		100	28	57	32					≤10	10000

STRAIN WAVE DRIVE REDUCERS

HD Mini-03

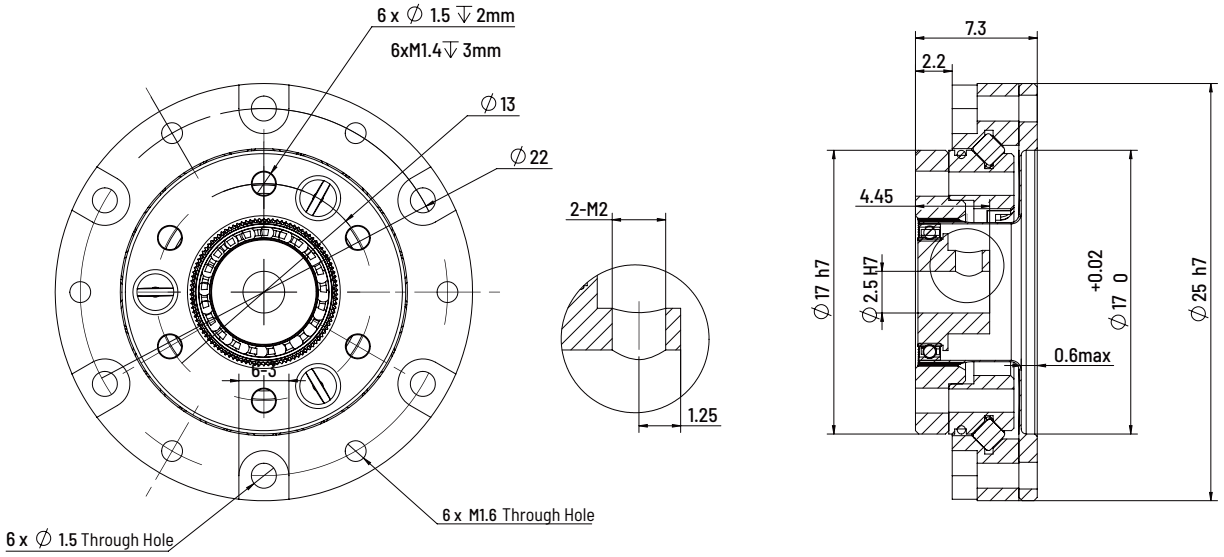
Strain Wave Gear



Diameter: 25mm | Thickness: 7.3mm | Weight: 0.016kg

Drawing

unit: mm



HD-03 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	0.08	0.13			
Limit for Repeated Peak Torque[Nm]	0.15	0.24			
Limit for Average Torque[Nm]	0.12	0.16			
Limit for Momenta. Peak Torque[Nm]	0.25	0.5			
Maximum Input Speed[rpm]	10000	10000			
Limit for Average Input Speed[rpm]	6500	6500			
Moment of Inertia [Ix10^-4 kgm^2]					
Backlash[Arcsec]	≤20	≤20			

STRAIN WAVE DRIVE REDUCERS

HD Mini-05

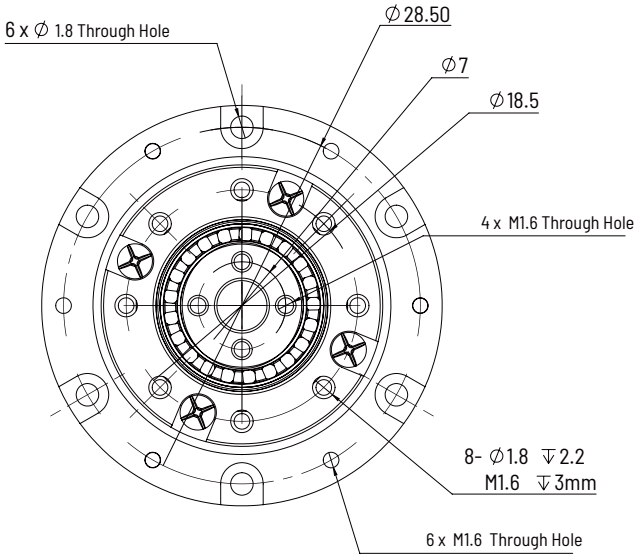
Strain Wave Gear

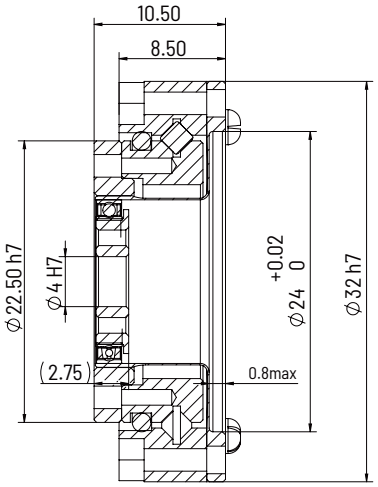


Diameter: 32mm | Thickness: 10.5mm | Weight: ≤0.04kg

Drawing

unit: mm





HD-05 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	0.23	0.38	0.5		
Limit for Repeated Peak Torque[Nm]	0.47	0.87	1.2		
Limit for Average Torque[Nm]	0.35	0.5	0.82		
Limit for Momentary Peak Torque[Nm]	0.85	1.7	2.2		
Maximum Input Speed[rpm]	10000	10000	10000		
Limit for Average Input Speed[rpm]	6500	6500	6500		
Moment of Inertia [Ix10^-4 kgm^2]					
Backlash[Arcsec]	≤20	≤20	≤20		

STRAIN WAVE DRIVE REDUCERS

HD Mini-08

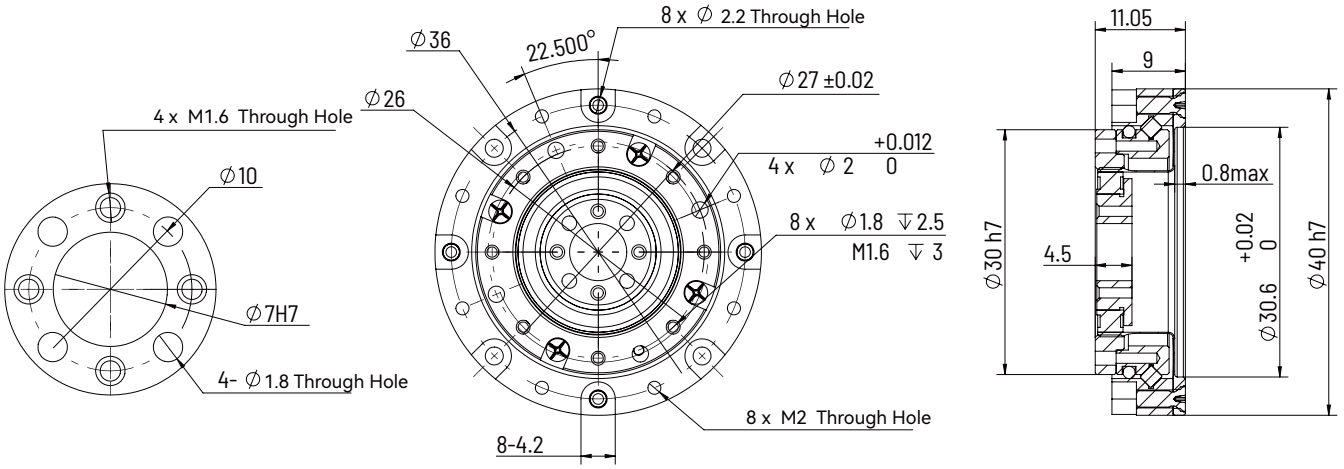
Strain Wave Gear



Diameter: 40mm | Thickness: 11.5mm | Weight: ≤0.061kg

Drawing

unit: mm



HD-08 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	0.8	1.6		2.4	
Limit for Repeated Peak Torque[Nm]	1.6	3.1		4.6	
Limit for Average Torque[Nm]	1.3	2.1		3.2	
Limit for Momentary Peak Torque[Nm]	3.2	6.1		8.7	
Maximum Input Speed[rpm]	8500	8500		8500	
Limit for Average Input Speed[rpm]	3500	3500		3500	
Moment of Inertia [Ix10^-4 kgm^2]					
Backlash[Arcsec]	≤20	≤20		≤20	

STRAIN WAVE DRIVE REDUCERS

HD Mini-11

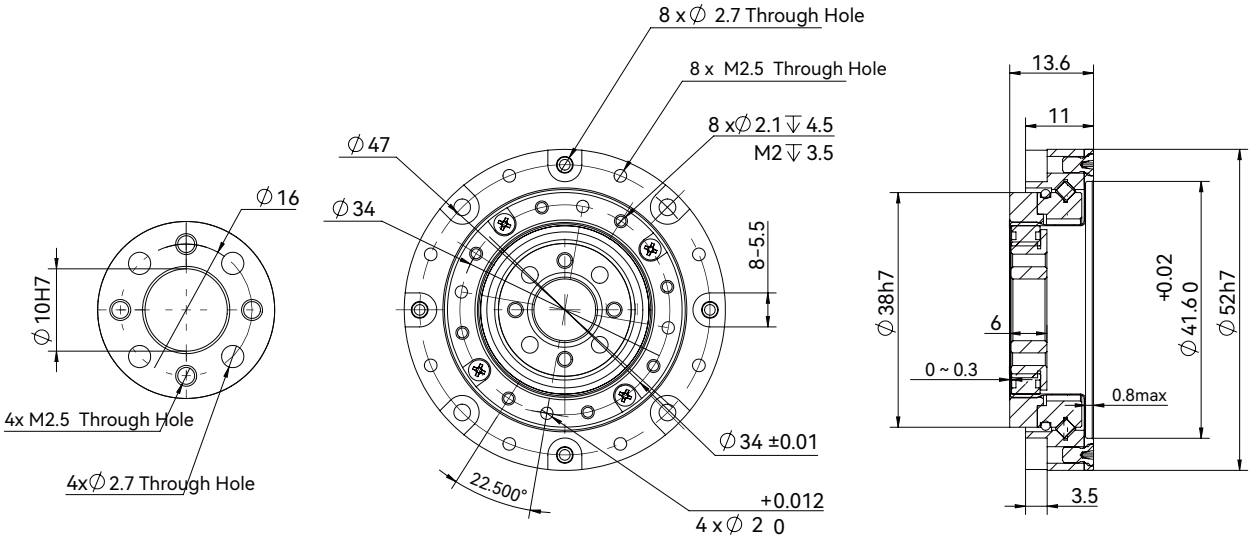
Strain Wave Gear



Diameter: 52mm | Thickness: 13.5mm | Weight: ≤0.127kg

Drawing

unit: mm



HD-11 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	2.1	3.2		4.8	
Limit for Repeated Peak Torque[Nm]	4.1	7.8		10.5	
Limit for Average Torque[Nm]	3.1	5.1		8.7	
Limit for Momentary Peak Torque[Nm]	8.2	15		23	
Maximum Input Speed[rpm]	8500	8500		8500	
Limit for Average Input Speed[rpm]	3500	3500		3500	
Moment of Inertia [Ix10^-4 kgm^2]					
Backlash[Arcsec]	≤20	≤20		≤20	

STRAIN WAVE DRIVE REDUCERS

HD-14

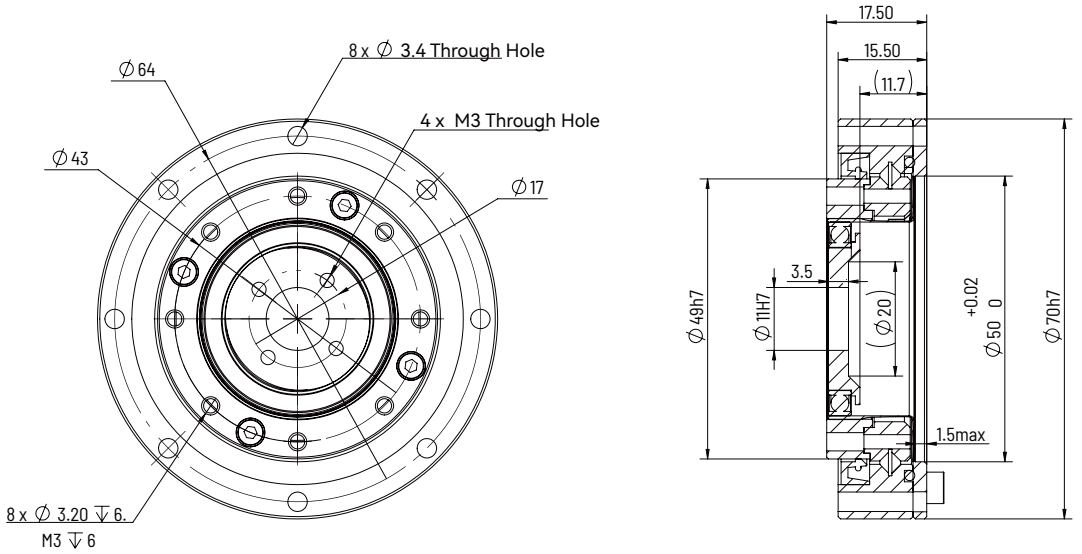
Strain Wave Gear



Diameter: 70mm | Thickness: 17.5mm | Weight: ≤0.312kg

Drawing

unit: mm



HD-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	3	4.5	6	7.4	
Limit for Repeated Peak Torque[Nm]	8.7	13	15	19	
Limit for Average Torque[Nm]	6.2	6.8	9.5	10.8	
Limit for Momentary Peak Torque[Nm]	15	25	38	40	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.021	0.021	0.021	0.021	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HD-17

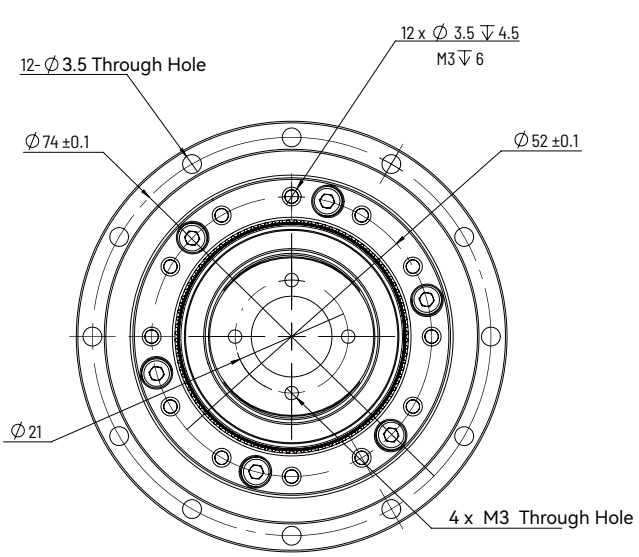
Strain Wave Gear

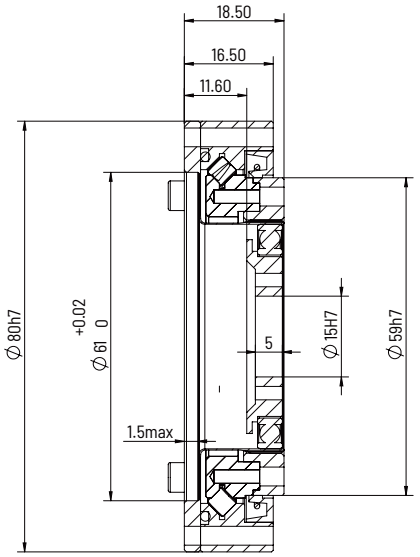


Diameter: 80mm | Thickness: 18.5mm | Weight: ≤0.4kg

Drawing

unit: mm





HD-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	6.6	11	12.8	16	
Limit for Repeated Peak Torque[Nm]	13.8	23	29.6	37	
Limit for Average Torque[Nm]	10.8	18	21.6	27.8	
Limit for Momentary Peak Torque[Nm]	28.8	48	56.8	71	
Maximum Input Speed[rpm]	7000	7000	7000	7000	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.054	0.054	0.054	0.054	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HD-20

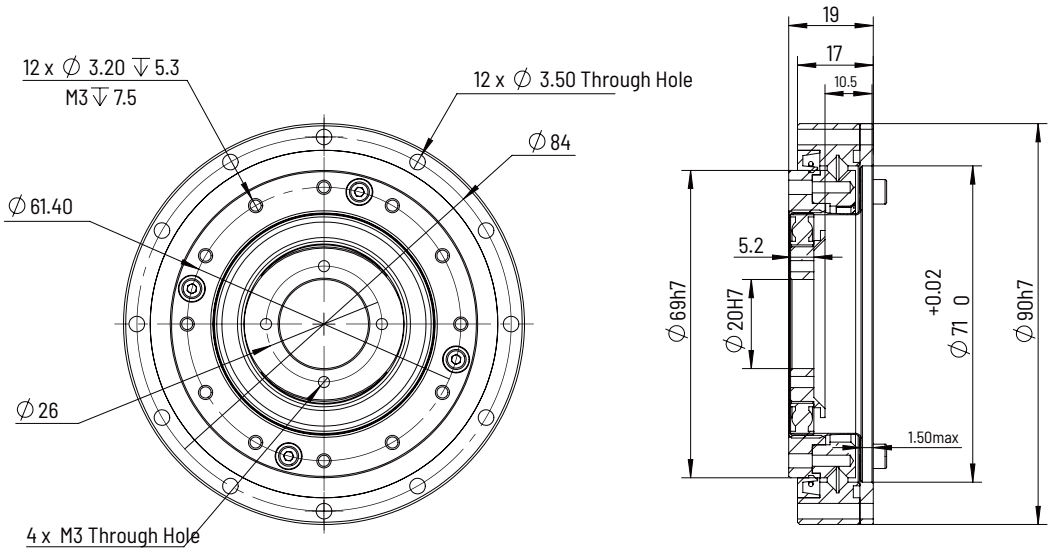
Strain Wave Gear



Diameter: 90mm | Thickness: 19mm | Weight: ≤0.513kg

Drawing

unit: mm



HD-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	10.2	17	22.4	28	28
Limit for Repeated Peak Torque[Nm]	23.4	39	45.6	57	61
Limit for Average Torque[Nm]	14.4	24	27.2	32	35
Limit for Momentary Peak Torque[Nm]	41.4	69	76	95	95
Maximum Input Speed[rpm]	7000	7000	7000	7000	7000
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.09	0.09	0.09	0.09	0.09
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HD-25

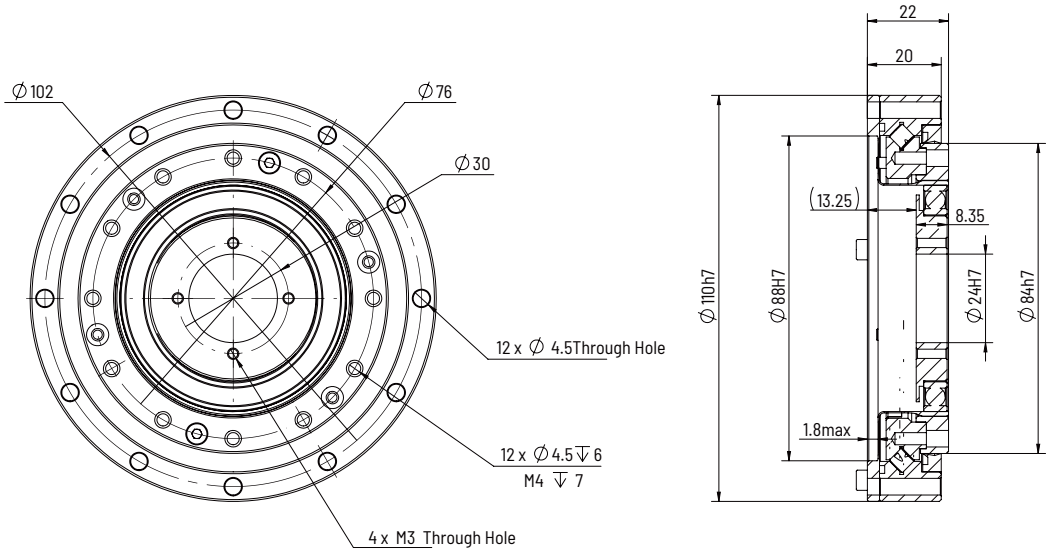
Strain Wave Gear



Diameter: 110mm | Thickness: 22mm | Weight: ≤0.926kg

Drawing

unit: mm



HD-25 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	16.2	26	44	48	54
Limit for Repeated Peak Torque[Nm]	41.4	69	95	110	120
Limit for Average Torque[Nm]	22.8	38	60	75	80
Limit for Momentary Peak Torque[Nm]	76.2	127	147.2	184	202
Maximum Input Speed[rpm]	5600	5600	5600	5600	5600
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.282	0.282	0.282	0.282	0.282
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HD-11 Pro

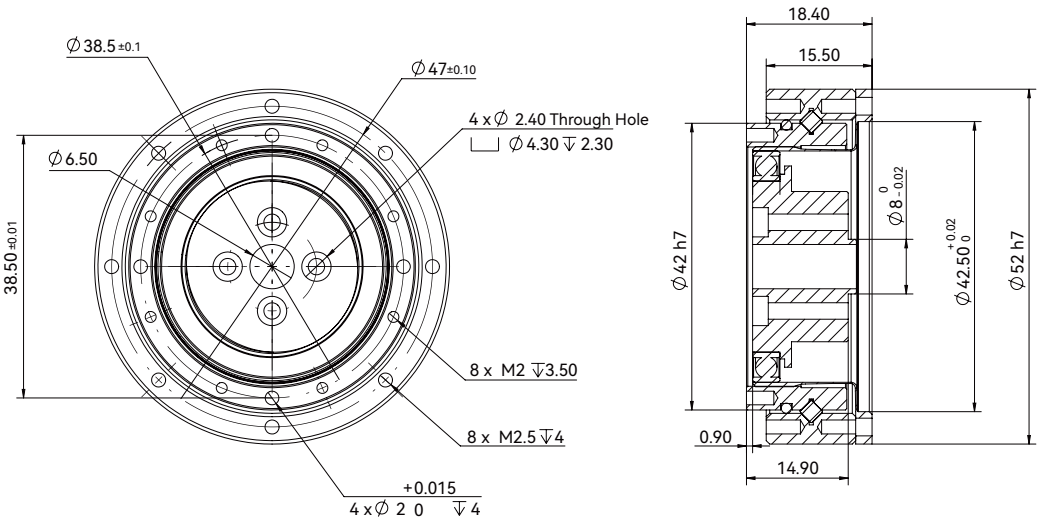
Strain Wave Gear



Diameter: 52mm | Thickness: 18.4mm | Weight: 0.17kg

Drawing

unit: mm



HD-11 Pro [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		4.5		7.4	
Limit for Repeated Peak Torque[Nm]		13		19	
Limit for Average Torque[Nm]		6.8		10.8	
Limit for Momentary Peak Torque[Nm]		25		40	
Maximum Input Speed[rpm]		8500		8500	
Limit for Average Input Speed[rpm]		3500		3500	
Moment of Inertia [Ix10^-4 kgm^2]		0.021		0.021	
Backlash[Arcsec]		≤20		≤10	

STRAIN WAVE DRIVE REDUCERS

HD-14 Pro

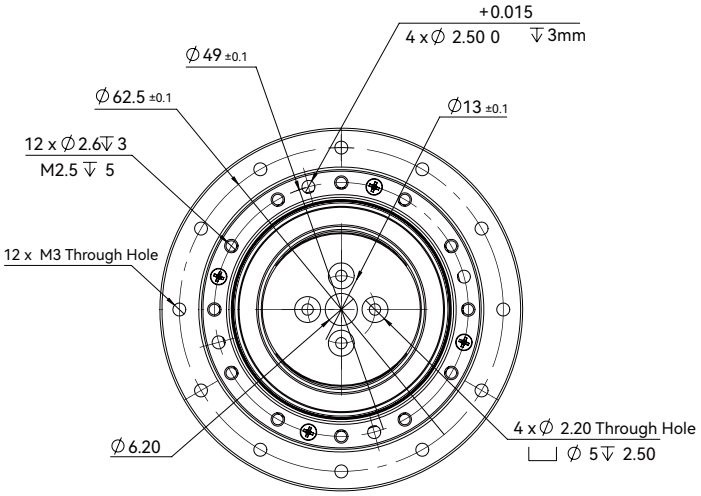
Strain Wave Gear

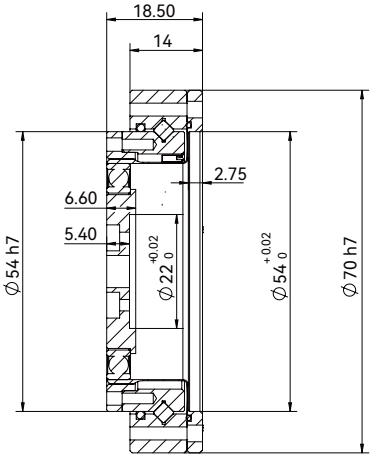


Diameter: 70mm | Thickness: 18.5mm | Weight: 0.278kg

Drawing

unit: mm





HD-14 Pro [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		11		16	
Limit for Repeated Peak Torque[Nm]		23		37	
Limit for Average Torque[Nm]		18		27.8	
Limit for Momentary Peak Torque[Nm]		48		71	
Maximum Input Speed[rpm]		7000		7000	
Limit for Average Input Speed[rpm]		3500		3500	
Moment of Inertia [Ix10^-4 kgm^2]		0.054		0.054	
Backlash[Arcsec]		≤20		≤10	

STRAIN WAVE DRIVE REDUCERS

HD-17 Pro

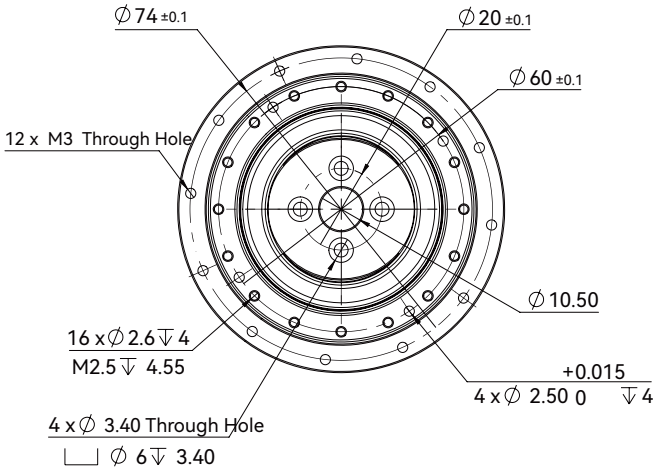
Strain Wave Gear

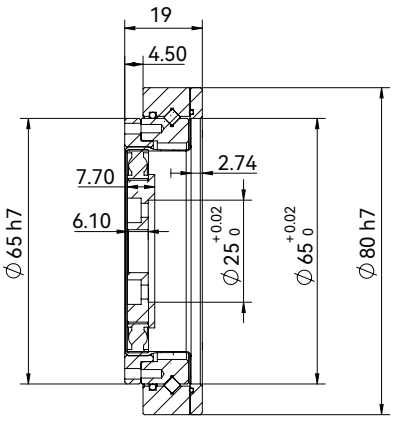


Diameter: 80mm | Thickness: 18.5mm | Weight: 0.383kg

Drawing

unit: mm





HD-17 Pro [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		17		28	
Limit for Repeated Peak Torque[Nm]		39		57	
Limit for Average Torque[Nm]		24		32	
Limit for Momentary Peak Torque[Nm]		69		95	
Maximum Input Speed[rpm]		7000		7000	
Limit for Average Input Speed[rpm]		3500		3500	
Moment of Inertia [Ix10^-4 kgm^2]		0.09		0.09	
Backlash[Arcsec]		≤20		≤10	



HD SERIES PRODUCTS

Strain Wave Drive Reducers

HS

STRAIN WAVE DRIVE REDUCERS

HS Series

Model	Ratio	Page
HS-14	30/50/80/100	35
HS-17	30/50/80/100	39
HS-20	30/50/80/100/120	43
HS-25	30/50/80/100/120	47
HS-32	50/80/100/120	51

I Simplicity unit
II Hollow shaft
III Standard
IV Input shaft



Features and Applications

The HS series is a modular product centered around ease of operation. It features a precision and high-rigidity cross roller bearing, which directly suppo. s the external load of the main bearing.
The application fields of the HS series include industrial automation, medical equipment, precision instruments, and other areas.

Features

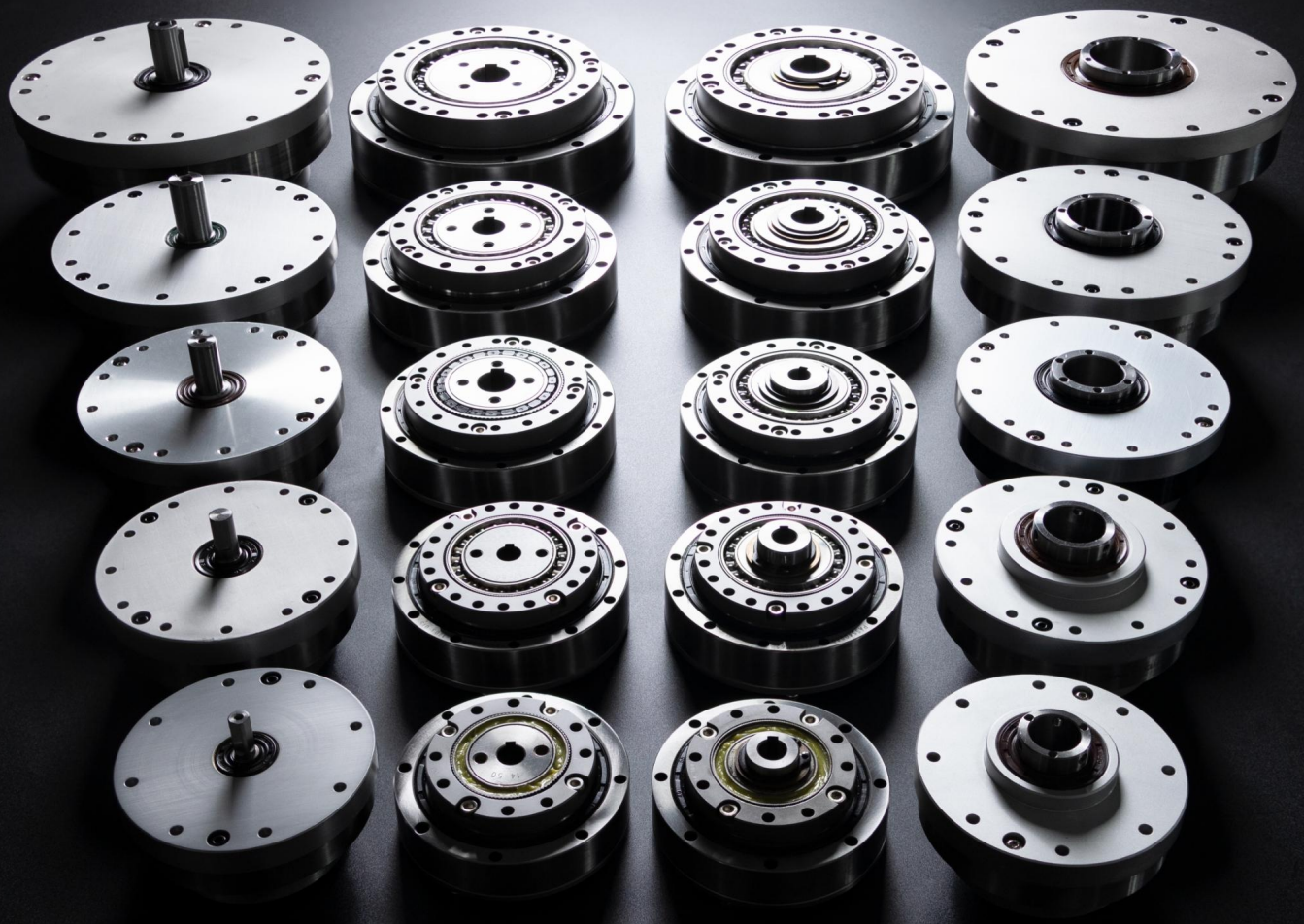
- Input and Output Coaxial
- Compact
- Large Hollow Sharft
- High Rigidity
- High Torque Capacity
- Excellent Positioning Accuracy

Product Model Description

HS-20-IV-100

Series	Model	Type	Gear Ratio						-	-
HS	14	I Simplicity unit II Hollow Sha t III Standard IV Input Sha t	30	50	80	100			-	-
	17		30	50	80	100				
	20		30	50	80	100	120			
	25		30	50	80	100	120			
	32			50	80	100	120			

Series	Model	Ratio	Rate Torque at 2000 rpm [N.m]	Limit for Repeated Peak Torque [N.m]	Limit for Average Torque [N.m]	Limit for Momentary Peak Torque [N.m]	Maximum Input Speed [r/min]	Limit for Average Input Speed [r/min]	Moment of Inertia ($1/4GD^2$) [$kgcm^2$]	Backlash [Arc sec]	Life [hour]
HS	HS-14	30	4	9	6.8	17	8500	3500	0.033	≤20	7000
		50	5.4	18	6.9	35					
		80	7.8	23	11	47					10000
		100	7.8	28	11	54					
	HS-17	30	8.8	16	12	30	7300	3500	0.079	≤20	7000
		50	16	34	26	70					
		80	22	43	27	87				≤10	10000
		100	24	54	39	110					
	HS-20	30	15	27	20	50	6500	3500	0.193	≤20	7000
		50	25	56	34	98					
		80	34	74	47	127				≤10	10000
		100	40	82	49	147					
		120	40	87	49	147					
	HS-25	30	27	50	38	95	5600	3500	0.413	≤20	7000
		50	39	98	55	186					
		80	63	137	87	255				≤10	10000
		100	67	157	108	284					
		120	67	167	108	304					
	HS-32	50	76	216	108	382	4800	3500	1.69	≤20	7000
		80	118	304	167	568					
		100	137	333	216	647				≤10	10000
		120	137	353	216	686					



HS SERIES PRODUCTS

Strain Wave Drive Reducers

STRAIN WAVE DRIVE REDUCERS

HS-14-I

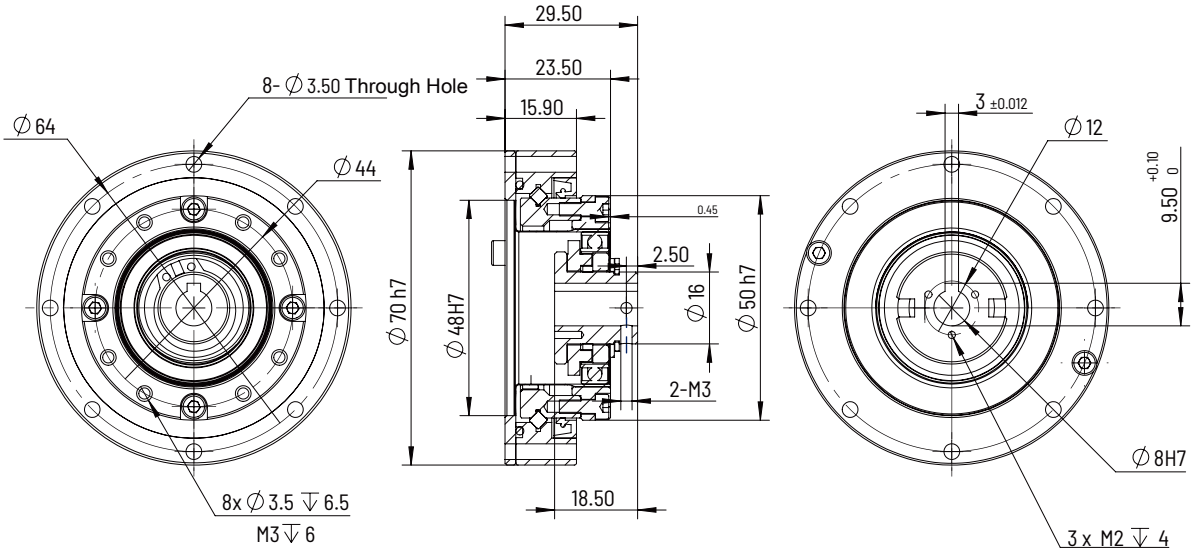
Simplicity Unit Strain Wave Gear



Diameter: Φ70mm | Thickness: 29.5mm | Weight≤0.374kg

Drawing

unit: mm



HS-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	4	5.4	7.8	7.8	
Limit for Repeated Peak Torque[Nm]	9	18	23	28	
Limit for Average Torque[Nm]	6.8	6.9	11	11	
Limit for Momenta. Peak Torque[Nm]	17	35	47	54	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.033	0.033	0.033	0.033	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-14-II

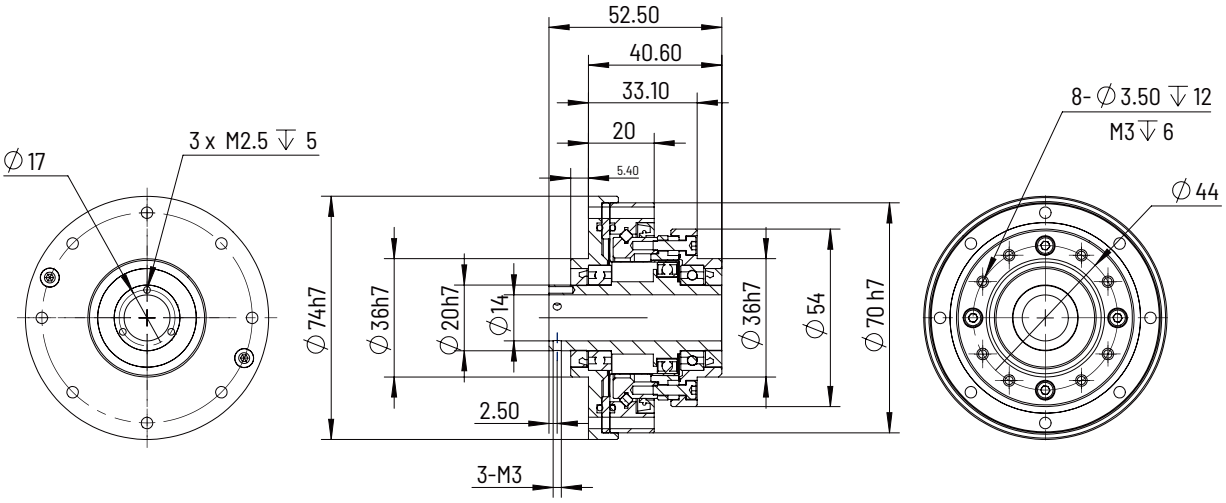
Hollow Sha t Strain Wave Gear



Diameter: Φ74mm | Thickness: 52.6mm | Weight≤0.55kg

Drawing

unit: mm



HS-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	4	5.4	7.8	7.8	
Limit for Repeated Peak Torque[Nm]	9	18	23	28	
Limit for Average Torque[Nm]	6.8	6.9	11	11	
Limit for Momentary Peak Torque[Nm]	17	35	47	54	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.033	0.033	0.033	0.033	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-14-III

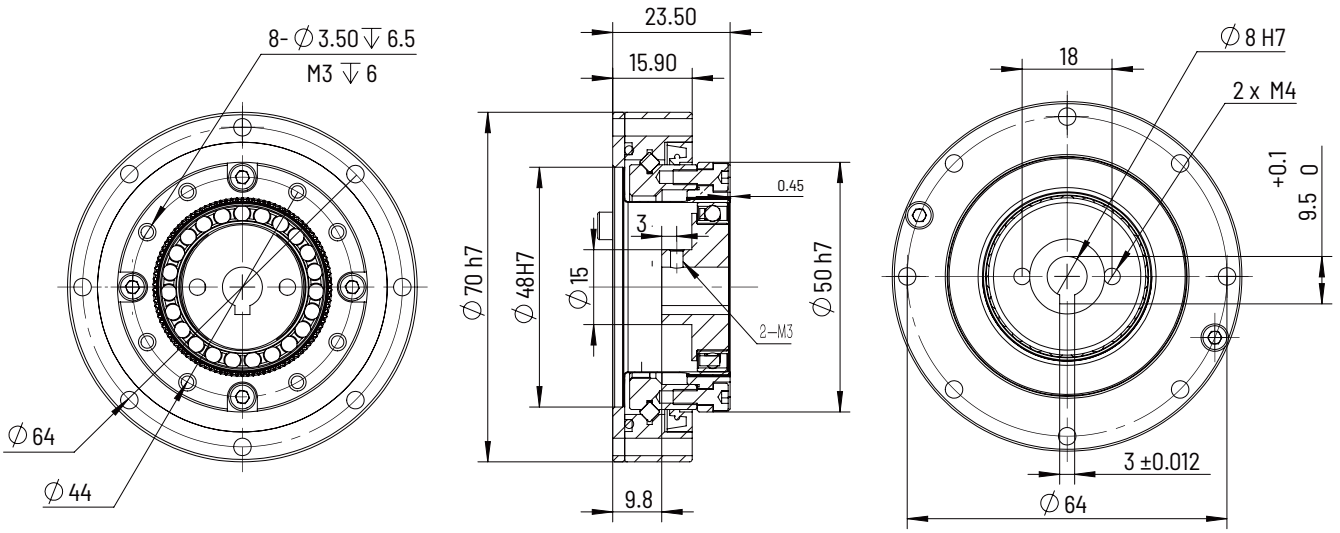
Standard Strain Wave Gear



Diameter: Φ70mm | Thickness: 23.5mm | Weight≤0.366kg

Drawing

unit: mm



HS-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	4	5.4	7.8	7.8	
Limit for Repeated Peak Torque[Nm]	9	18	23	28	
Limit for Average Torque[Nm]	6.8	6.9	11	11	
Limit for Momentary Peak Torque[Nm]	17	35	47	54	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.033	0.033	0.033	0.033	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-14-IV

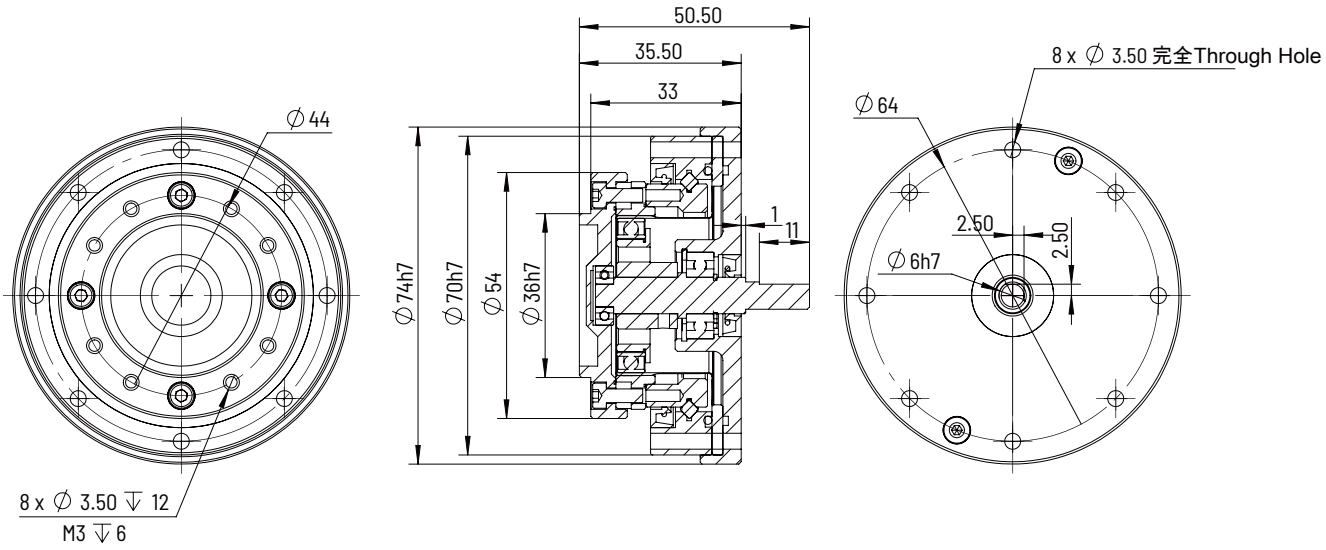
Input Sha t Strain Wave Gear



Diameter: Φ74mm | Thickness: 50.5mm | Weight≤0.471kg

Drawing

unit: mm



HS-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	4	5.4	7.8	7.8	
Limit for Repeated Peak Torque[Nm]	9	18	23	28	
Limit for Average Torque[Nm]	6.8	6.9	11	11	
Limit for Momentary Peak Torque[Nm]	17	35	47	54	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.033	0.033	0.033	0.033	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-17-I

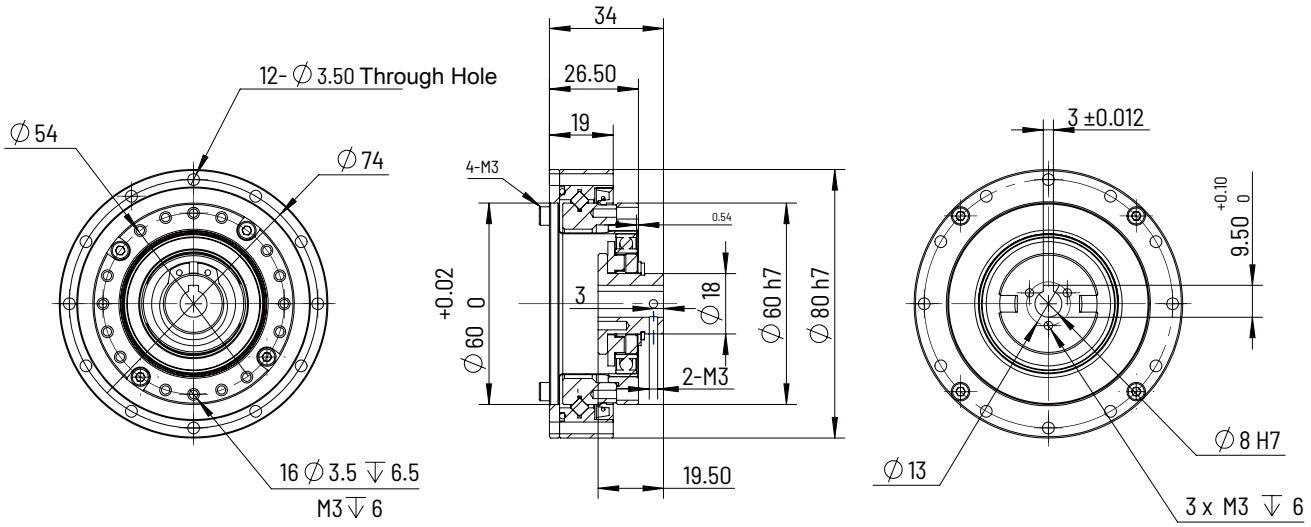


Simplicity Unit Strain Wave Gear

Diameter: Φ80mm | Thickness: 34mm | Weight≤0.55kg

Drawing

unit: mm



HS-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	8.8	16	22	24	
Limit for Repeated Peak Torque[Nm]	16	34	43	54	
Limit for Average Torque[Nm]	12	26	27	39	
Limit for Momentary Peak Torque[Nm]	30	70	87	110	
Maximum Input Speed[rpm]	7300	7300	7300	7300	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.079	0.079	0.079	0.079	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-17-II

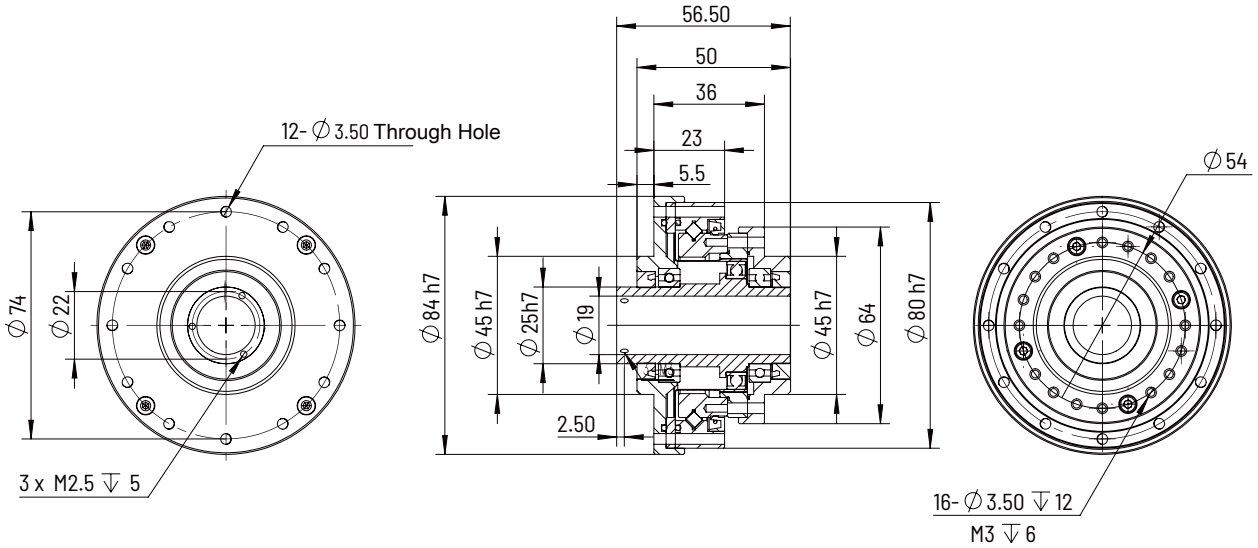
Hollow Sha t Strain Wave Gear



Diameter: Φ84mm | Thickness: 56.6mm | Weight≤0.763kg

Drawing

unit: mm



HS-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	8.8	16	22	24	
Limit for Repeated Peak Torque[Nm]	16	34	43	54	
Limit for Average Torque[Nm]	12	26	27	39	
Limit for Momentary Peak Torque[Nm]	30	70	87	110	
Maximum Input Speed[rpm]	7300	7300	7300	7300	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.079	0.079	0.079	0.079	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-17-III

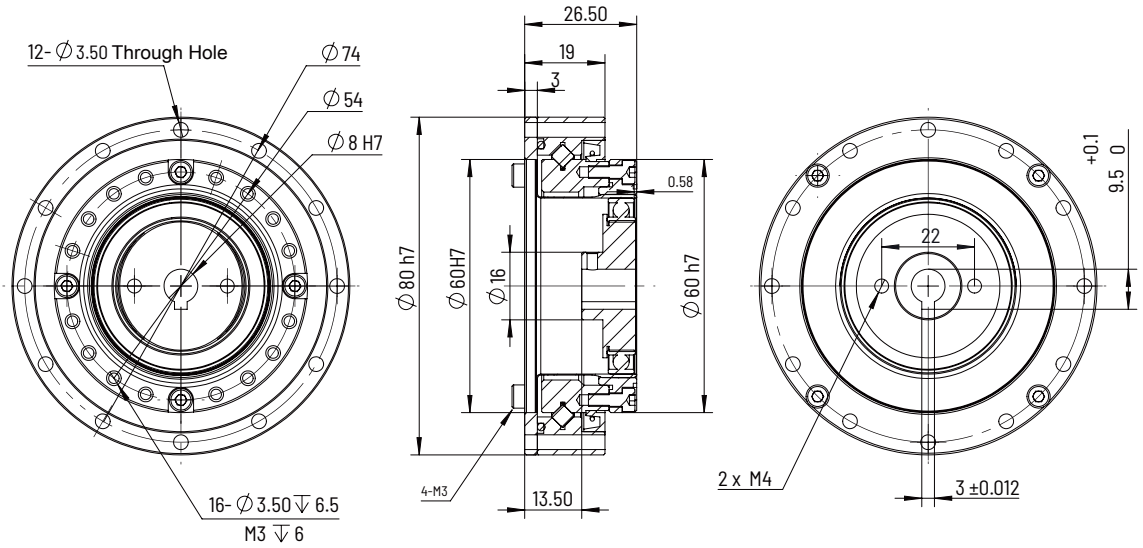
Standard Strain Wave Gear



Diameter: Φ80mm | Thickness: 26.5mm | Weight≤0.53kg

Drawing

unit: mm



HS-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	8.8	16	22	24	
Limit for Repeated Peak Torque[Nm]	16	34	43	54	
Limit for Average Torque[Nm]	12	26	27	39	
Limit for Momentary Peak Torque[Nm]	30	70	87	110	
Maximum Input Speed[rpm]	7300	7300	7300	7300	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.079	0.079	0.079	0.079	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-17-IV

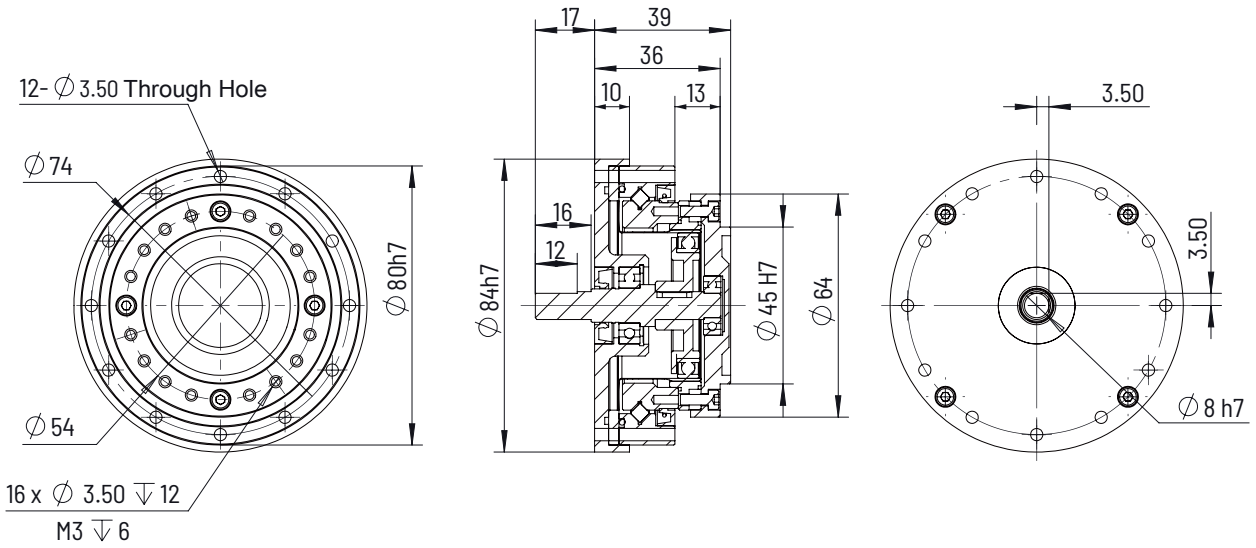
Input Sha t Strain Wave Gear



Diameter: Φ84mm | Thickness: 55mm | Weight≤0.674kg

Drawing

unit: mm



HS-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	8.8	16	22	24	
Limit for Repeated Peak Torque[Nm]	16	34	43	54	
Limit for Average Torque[Nm]	12	26	27	39	
Limit for Momentary Peak Torque[Nm]	30	70	87	110	
Maximum Input Speed[rpm]	7300	7300	7300	7300	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.079	0.079	0.079	0.079	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

HS-20-I

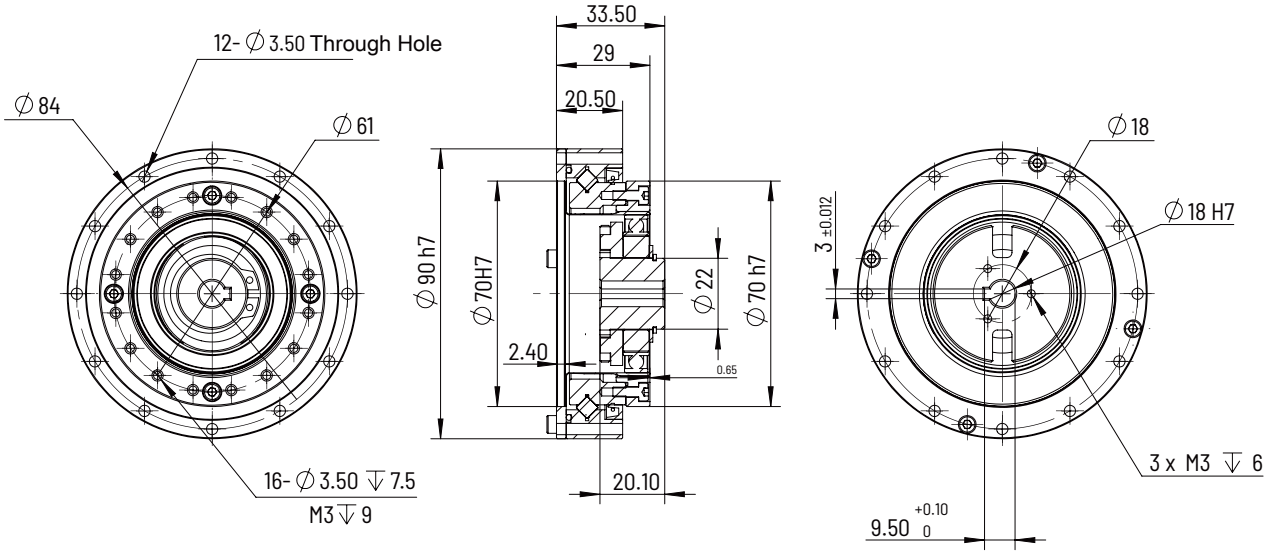


Simplicity Unit Strain Wave Gear

Diameter: Φ90mm | Thickness: 33.5mm | Weight≤0.787kg

Drawing

unit: mm



HS-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	15	25	34	40	40
Limit for Repeated Peak Torque[Nm]	27	56	74	82	87
Limit for Average Torque[Nm]	20	34	47	49	49
Limit for Momentary Peak Torque[Nm]	50	98	127	147	147
Maximum Input Speed[rpm]	6500	6500	6500	6500	6500
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.193	0.193	0.193	0.193	0.193
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-20-II

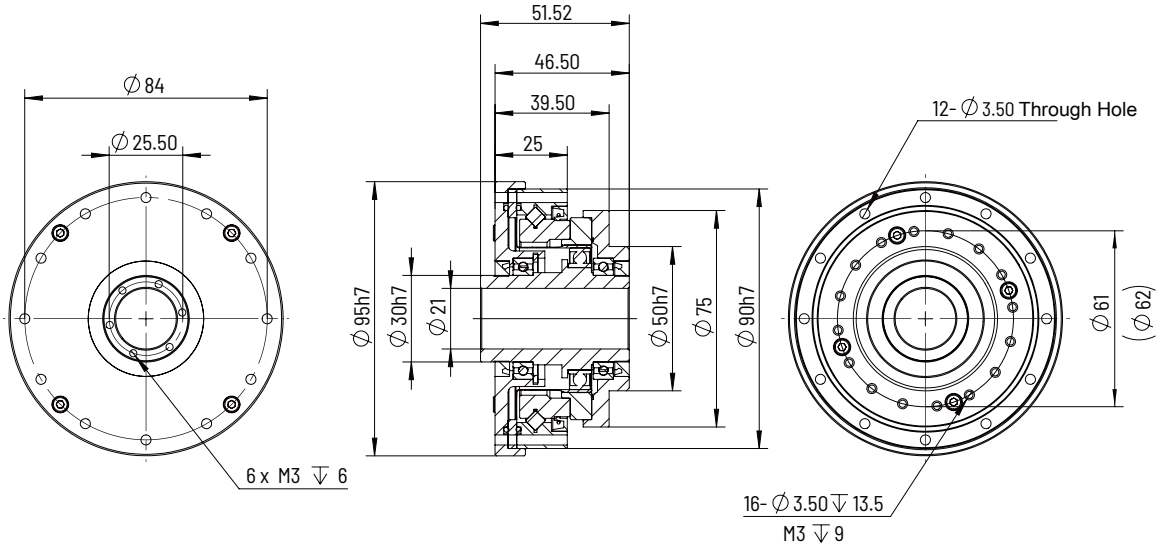
Hollow Sha t Strain Wave Gear



Diameter: Φ95mm | Thickness: 51.5mm | Weight≤1.061kg

Drawing

unit: mm



HS-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	15	25	34	40	40
Limit for Repeated Peak Torque[Nm]	27	56	74	82	87
Limit for Average Torque[Nm]	20	34	47	49	49
Limit for Momentary Peak Torque[Nm]	50	98	127	147	147
Maximum Input Speed[rpm]	6500	6500	6500	6500	6500
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.193	0.193	0.193	0.193	0.193
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-20-III

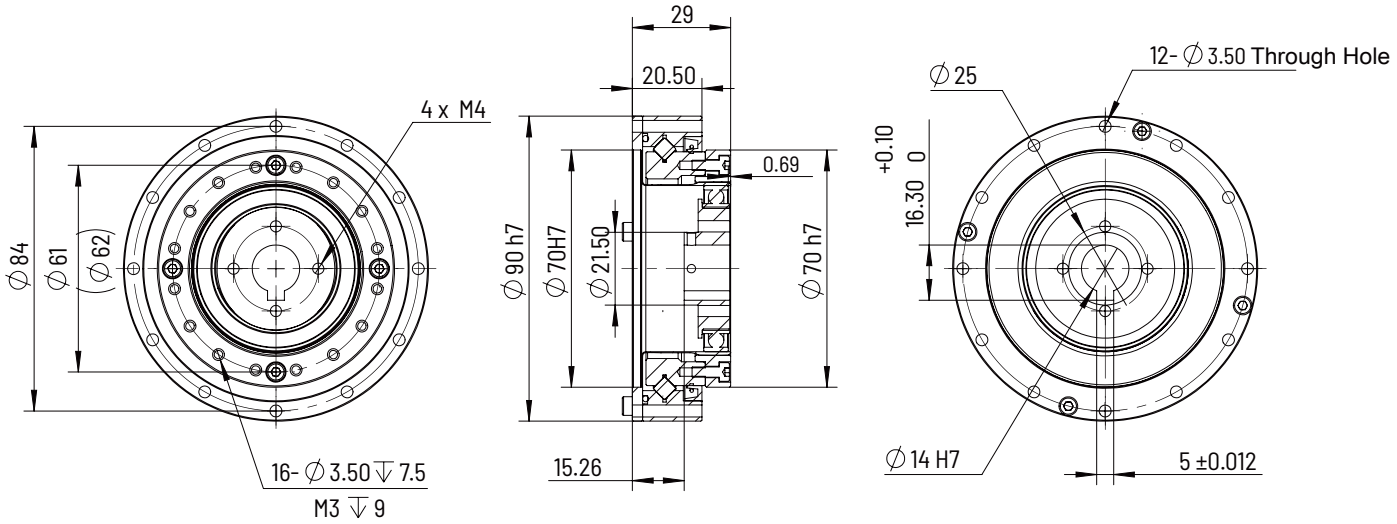
Standard Strain Wave Gear



Diameter: Φ90mm | Thickness: 29mm | Weight≤0.75kg

Drawing

unit: mm



HS-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	15	25	34	40	40
Limit for Repeated Peak Torque[Nm]	27	56	74	82	87
Limit for Average Torque[Nm]	20	34	47	49	49
Limit for Momentary Peak Torque[Nm]	50	98	127	147	147
Maximum Input Speed[rpm]	6500	6500	6500	6500	6500
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.193	0.193	0.193	0.193	0.193
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-20-IV

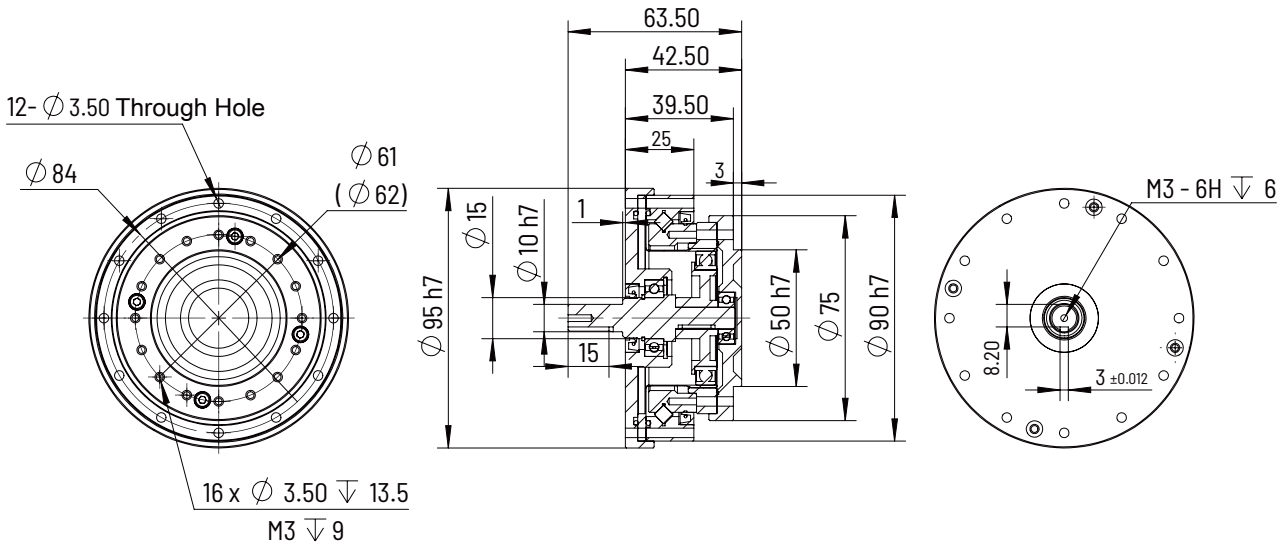
Input Sha t Strain Wave Gear



Diameter: Φ95mm | Thickness: 63.5mm | Weight≤0.965kg

Drawing

unit: mm



HS-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	15	25	34	40	40
Limit for Repeated Peak Torque[Nm]	27	56	74	82	87
Limit for Average Torque[Nm]	20	34	47	49	49
Limit for Momentary Peak Torque[Nm]	50	98	127	147	147
Maximum Input Speed[rpm]	6500	6500	6500	6500	6500
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.193	0.193	0.193	0.193	0.193
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-25-I

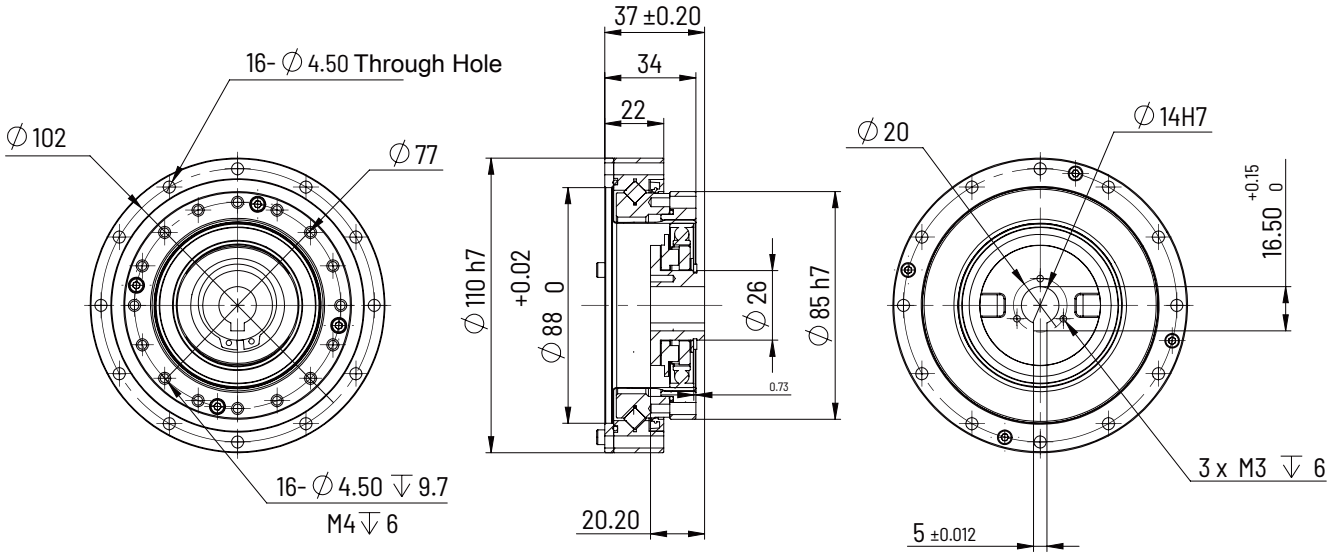
Simplicity Unit Strain Wave Gear



Diameter: Φ110mm | Thickness: 37.3mm | Weight≤1.25kg

Drawing

unit: mm



HS-25 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	27	39	63	67	67
Limit for Repeated Peak Torque[Nm]	50	98	137	157	167
Limit for Average Torque[Nm]	38	55	87	108	108
Limit for Momentary Peak Torque[Nm]	95	186	255	284	304
Maximum Input Speed[rpm]	5600	5600	5600	5600	5600
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.413	0.413	0.413	0.413	0.413
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-25-II

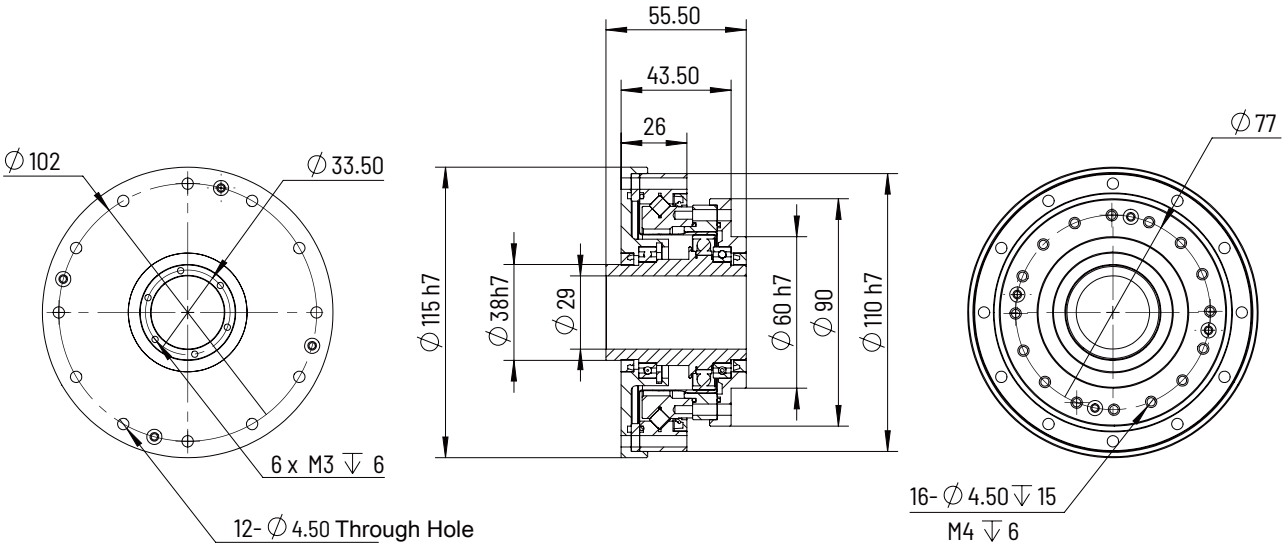
Hollow Sha t Strain Wave Gear



Diameter: Φ115mm | Thickness: 55.5mm | Weight≤1.663kg

Drawing

unit: mm



HS-25 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	27	39	63	67	67
Limit for Repeated Peak Torque[Nm]	50	98	137	157	167
Limit for Average Torque[Nm]	38	55	87	108	108
Limit for Momentary Peak Torque[Nm]	95	186	255	284	304
Maximum Input Speed[rpm]	5600	5600	5600	5600	5600
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.413	0.413	0.413	0.413	0.413
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-25-III

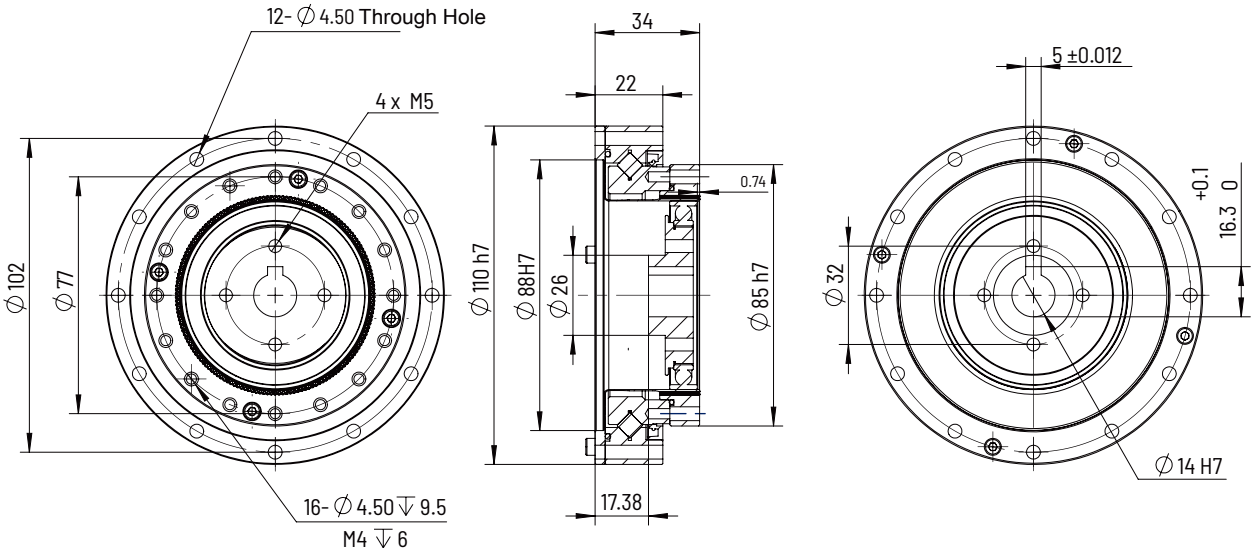
Standard Strain Wave Gear



Diameter: Φ110mm | Thickness: 34mm | Weight≤1.222kg

Drawing

unit: mm



HS-25 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	27	39	63	67	67
Limit for Repeated Peak Torque[Nm]	50	98	137	157	167
Limit for Average Torque[Nm]	38	55	87	108	108
Limit for Momentary Peak Torque[Nm]	95	186	255	284	304
Maximum Input Speed[rpm]	5600	5600	5600	5600	5600
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.413	0.413	0.413	0.413	0.413
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-25-IV

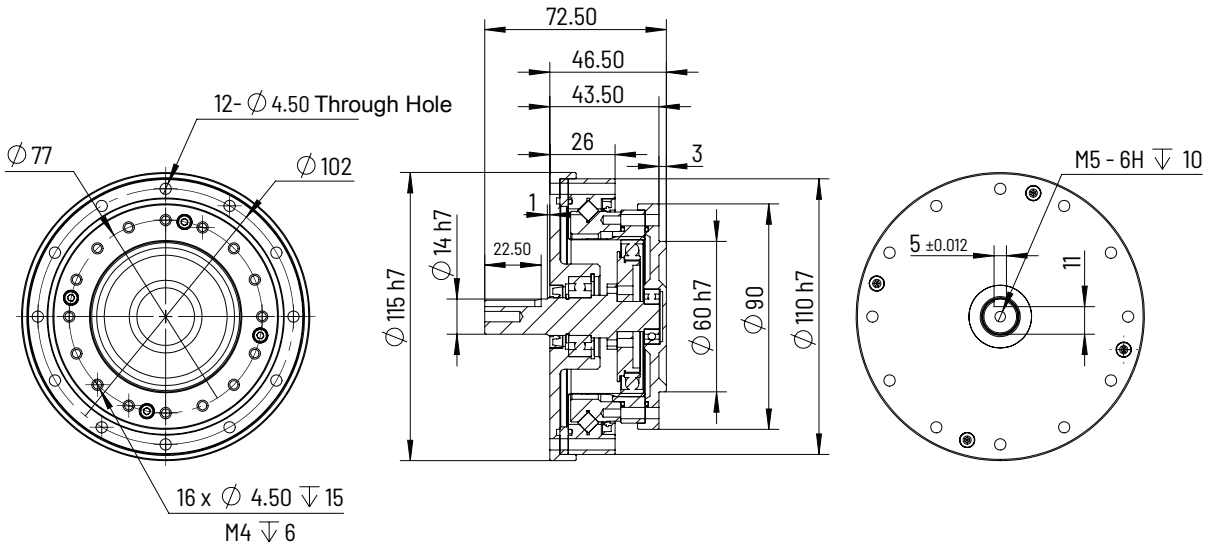


Input Sha t Strain Wave Gear

Diameter: Φ115mm | Thickness: 72.5mm | Weight≤1.556kg

Drawing

unit: mm



HS-25 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	27	39	63	67	67
Limit for Repeated Peak Torque[Nm]	50	98	137	157	167
Limit for Average Torque[Nm]	38	55	87	108	108
Limit for Momentary Peak Torque[Nm]	95	186	255	284	304
Maximum Input Speed[rpm]	5600	5600	5600	5600	5600
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.413	0.413	0.413	0.413	0.413
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-32-I

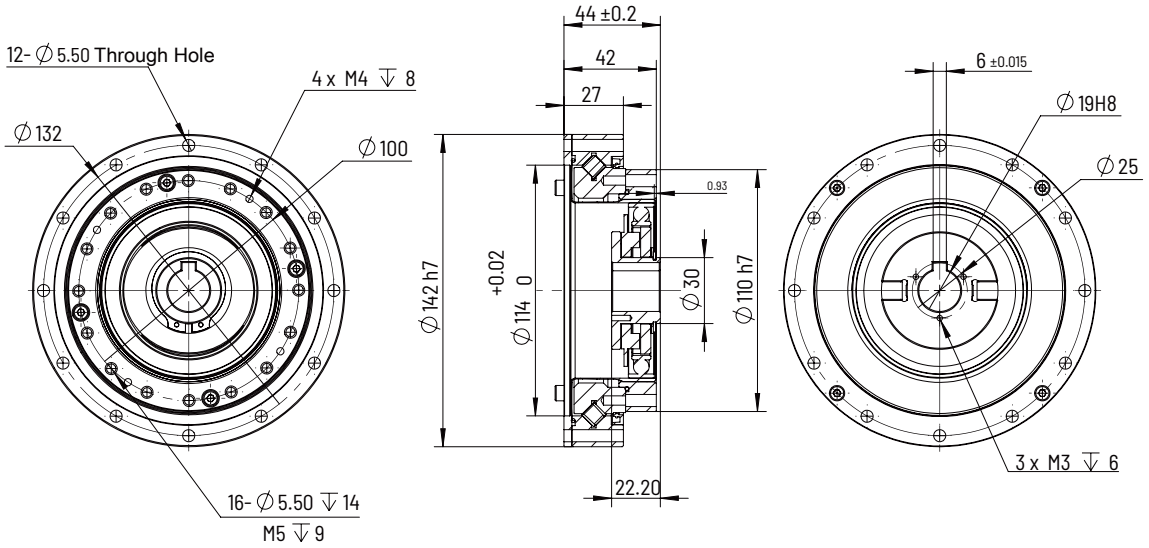


Simplicity Unit Strain Wave Gear

Diameter: Φ142mm | Thickness: 44mm | Weight≤2.6kg

Drawing

unit: mm



HS-32 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		76	118	137	137
Limit for Repeated Peak Torque[Nm]		216	304	333	353
Limit for Average Torque[Nm]		108	167	216	216
Limit for Momentary Peak Torque[Nm]		382	568	647	686
Maximum Input Speed[rpm]		4800	4800	4800	4800
Limit for Average Input Speed[rpm]		3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]		1.69	1.69	1.69	1.69
Backlash[Arcsec]		≤20	≤20	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-32-II

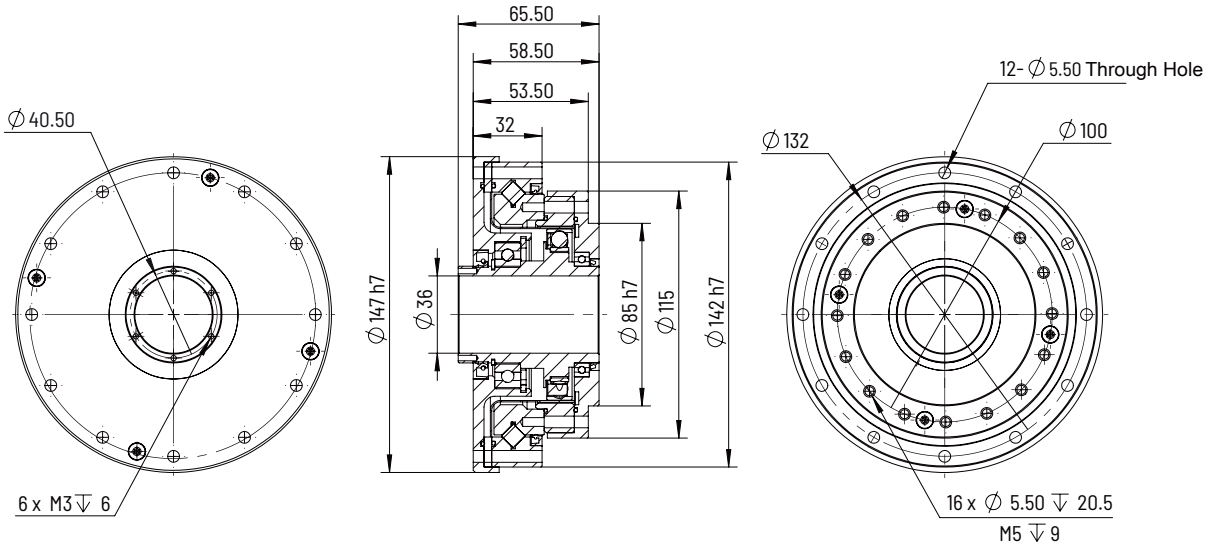
Hollow Sha t Strain Wave Gear



Diameter: Φ147mm | Thickness: 65.1mm | Weight≤3.411kg

Drawing

unit: mm



HS-32 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		76	118	137	137
Limit for Repeated Peak Torque[Nm]		216	304	333	353
Limit for Average Torque[Nm]		108	167	216	216
Limit for Momentary Peak Torque[Nm]		382	568	647	686
Maximum Input Speed[rpm]		4800	4800	4800	4800
Limit for Average Input Speed[rpm]		3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]		1.69	1.69	1.69	1.69
Backlash[Arcsec]		≤20	≤20	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-32-III

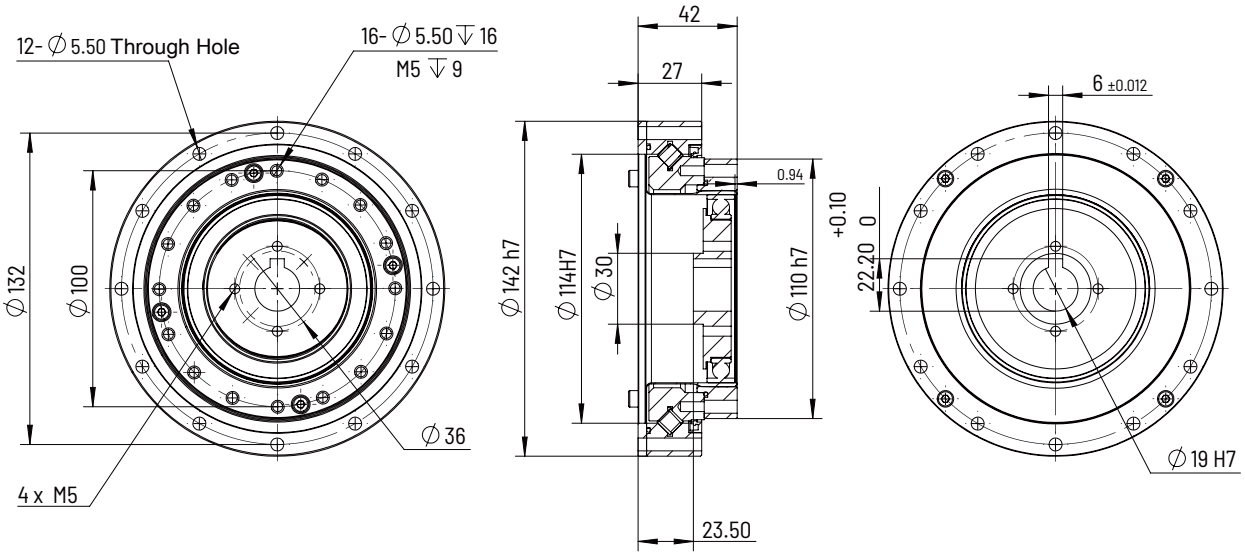
Standard Strain Wave Gear



Diameter: Φ142mm | Thickness: 42mm | Weight≤2.55kg

Drawing

unit: mm



HS-32 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		76	118	137	137
Limit for Repeated Peak Torque[Nm]		216	304	333	353
Limit for Average Torque[Nm]		108	167	216	216
Limit for Momentary Peak Torque[Nm]		382	568	647	686
Maximum Input Speed[rpm]		4800	4800	4800	4800
Limit for Average Input Speed[rpm]		3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]		1.69	1.69	1.69	1.69
Backlash[Arcsec]		≤20	≤20	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

HS-32-IV

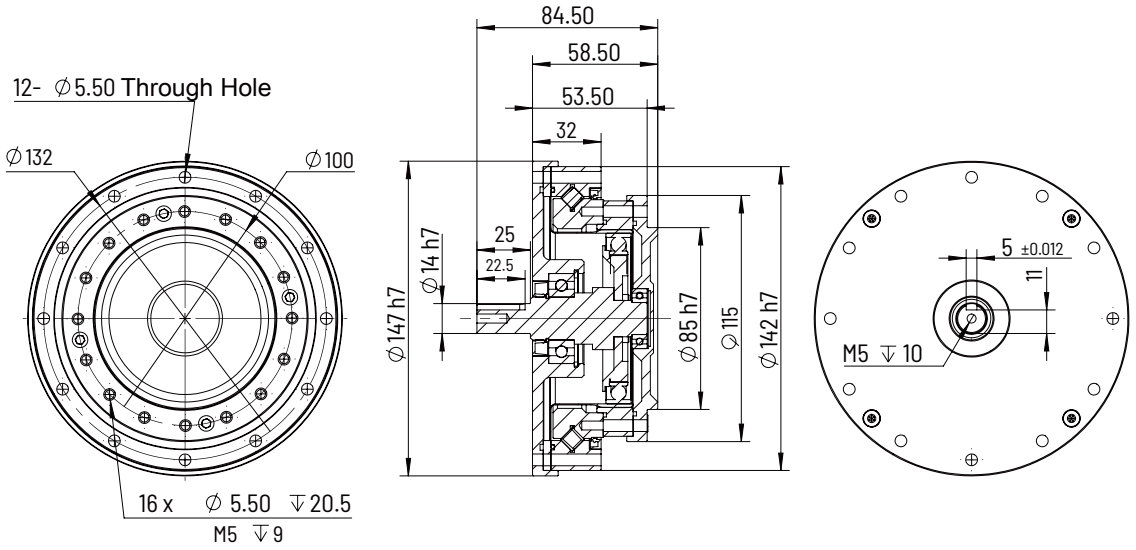
Input Sha t Strain Wave Gear



Diameter: Φ147mm | Thickness: 84.5mm | Weight≤3.171kg

Drawing

unit: mm



HS-32 [Parameter]

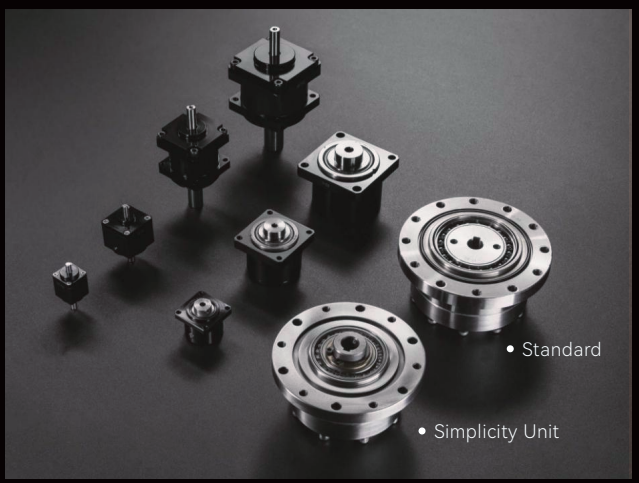
Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		76	118	137	137
Limit for Repeated Peak Torque[Nm]		216	304	333	353
Limit for Average Torque[Nm]		108	167	216	216
Limit for Momentary Peak Torque[Nm]		382	568	647	686
Maximum Input Speed[rpm]		4800	4800	4800	4800
Limit for Average Input Speed[rpm]		3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]		1.69	1.69	1.69	1.69
Backlash[Arcsec]		≤20	≤20	≤10	≤10

CS

STRAIN WAVE DRIVE REDUCERS

CS Series

Model	Gear Ratio	Page
CS-14	30/50/80/100	59
CS-17	30/50/80/100	61
CS-20	30/50/80/100/120	63
CS-25	30/50/80/100/120	65
CS-32	50/80/100/120	67
I Simplicity Unit II Standard		
CS mini-2.5	30/50	69
CS mini-03	30/50	72
CS mini-05	30/50/80	73
CS mini-08	30/50/100	75
CS mini-11	30/50/100	77



Features and Applications

The CS series represents a line of compact harmonic products, featuring a small-sized four-point contact ball bearing capable of directly suppo. ing external loads. This series is widely applied in precision instruments, small-scale automation equipment, and rotary receiving devices.

- High Torque Capacity
- High Rigidity
- Input and Output Coaxial
- Excellent Positioning Accuracy
- Compact

Product Model Description

CS-20-II-100

Series	Model	Type	Gear Ratio						-	-
CS	14	I Simplicity Unit II Standard	30	50	80	100			-	-
	17		30	50	80	100				
	20		30	50	80	100	120			
	25		30	50	80	100	120			
	32			50	80	100	120			
	mini 2.5		30	50						
	mini 03		30	50						
	mini 05		30	50	80					
	mini 08		30	50		100				
	mini 11		30	50		100				

Series	Model	Ratio	Rate Torque at 2000 rpm [N.m]	Limit for Repeated Peak Torque [N.m]	Limit for Average Torque [N.m]	Limit for Momentary Peak Torque [N.m]	Maximum Input Speed [r/min]	Limit for Average Input Speed [r/min]	Moment of Inertia ($1/4GD^2$) [$kgcm^2$]	Backlash [Arc sec]	Life [hour]
CS	CS-14	30	4	9	6.8	17	8500	3500	0.033	≤20	7000
		50	5.4	18	6.9	35					
		80	7.8	23	9.5	47				≤10	10000
		100	7.8	28	11	54					
	CS-17	30	8.8	16	12	30	7300	3500	0.079	≤20	7000
		50	16	34	26	70					
		80	22	43	27	87				≤10	10000
		100	24	54	39	108					
	CS-20	30	15	27	20	50	6500	3500	0.193	≤20	7000
		50	25	56	34	98					
		80	34	74	47	121				≤10	10000
		100	40	82	49	147					
		120	47	87	55	160					
	CS-25	30	27	50	38	95	5600	3500	0.413	≤20	7000
		50	39	98	55	186					
		80	63	137	87	255				≤10	10000
		100	67	157	108	289					
		120	78	167	115	304					
	CS-32	50	76	216	108	382	4800	3500	1.69	≤20	7000
		80	118	304	167	568					
		100	137	353	216	647				≤10	10000
		120	152	353	240	686					
CS Mini-2.5		30	0.06	0.13	0.1	0.21	7000	3500	/	≤20	7000
		50	0.10	0.21	0.13	0.40					
CS Mini	CS Mini-03	30	0.06	0.13	0.1	0.22	10000	6500	/	≤20	7000
		50	0.11	0.21	0.13	0.41					
	CS Mini-05	30	0.25	0.5	0.38	0.9	10000	6500	2.5x10^-4	≤20	7000
		50	0.4	0.9	0.53	1.8					
		80	0.52	1.35	0.84	2.5					10000
	CS Mini-08	30	0.9	1.8	1.4	3.3	8500	3500	3.2x10^-3	≤20	7000
		50	1.8	3.3	2.3	6.6					
		100	2.4	4.8	3.3	9					10000
	CS Mini-11	30	2.2	4.5	3.4	8.5	8500	3500	1.4x10^-2	≤20	7000
		50	3.5	8.3	5.5	17					
		100	5	11	8.9	25					10000

About Assembly

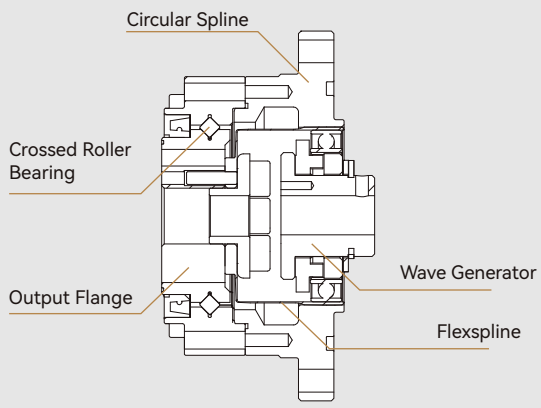
The CS-mini series is SITO's most compact and user-friendly modular product, capable of directly suppo ing external loads.
Please select the most suitable model based on the design requirements of your mechanical system.

Structure Type

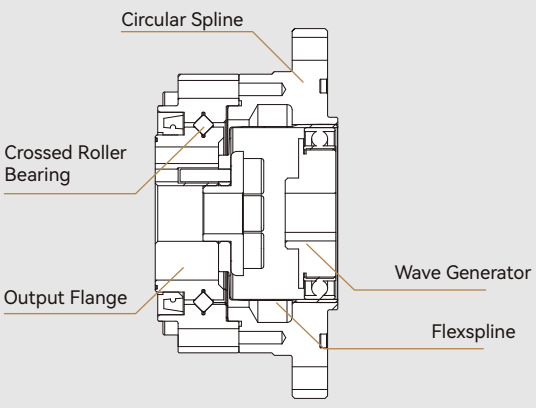
I Simplicity Unit **II** Standard

Mini Type (Refer to the illustration)

CS Series Structural Diagram

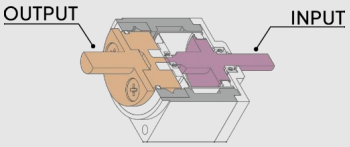


Simplicity Unit

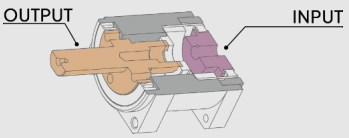


Standard

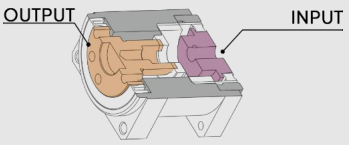
Input Sha t&Output Sha t
Related Type: **1U-U**



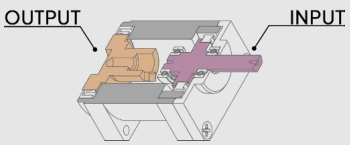
Input Hole & Output Sha t
Related Type: **1U-C**



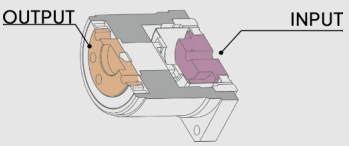
Input Hole & Output Flange
Related Type: **1U-D**



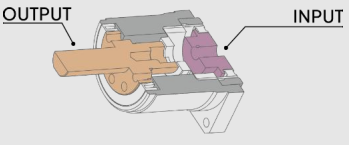
Input Sha t & Output Flange
Related Type: **1U-E**



Input Hole & Output Flange
Related Type: **2XH-A**



Input Hole & Output Sha t
Related Type: **2XH-B**



About the Sealing Mechanism

To prevent grease leakage and maintain the high durability of the reducer, the following sealing mechanisms must be used on the motor installation type:

Flange Assembly Su. ace ⟶	**O-ring, sealant: At this point, please pay attention to whether the surface is skewed and the engagement condition of the O-ring.**
Screw ⟶	**Use a sealing screw locking agent (Loctite 242 recommended) or sealing tape.**

About Lubrication

The standard lubrication method for the CS series is grease lubrication. The grease has been pre-
lled before leaving the factory, so there is no need to inject or apply grease during assembly.
Fu hermore, the following grease is used for lubrication.
If the grease volume/space volume exceeds 50%, grease leakage may occur. For such usage, please consult our company or an authorized distributor.

STRAIN WAVE DRIVE REDUCERS

CS-14-I

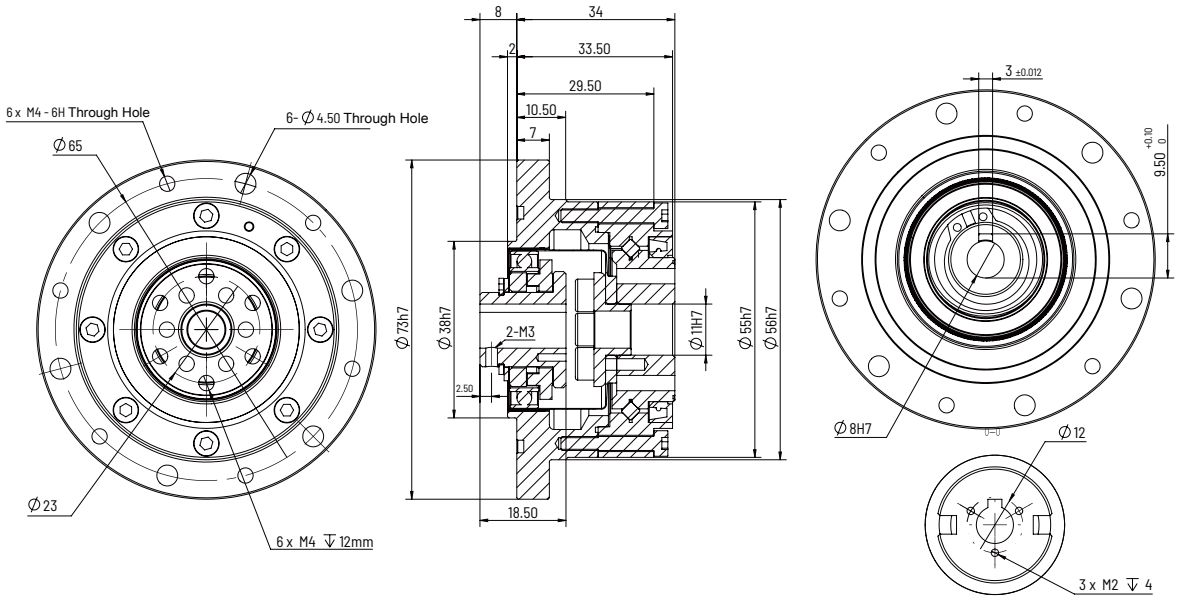


Simplicity Unit Strain Wave Gear

Diameter: Φ73mm | Thickness: 44.03mm | Weight≤0.531kg

Drawing

unit: mm



CS-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	4	5.4	7.8	7.8	
Limit for Repeated Peak Torque[Nm]	9	18	23	28	
Limit for Average Torque[Nm]	6.8	6.9	9.5	11	
Limit for Momentary Peak Torque[Nm]	17	35	47	54	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.033	0.033	0.033	0.033	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

CS-14-II

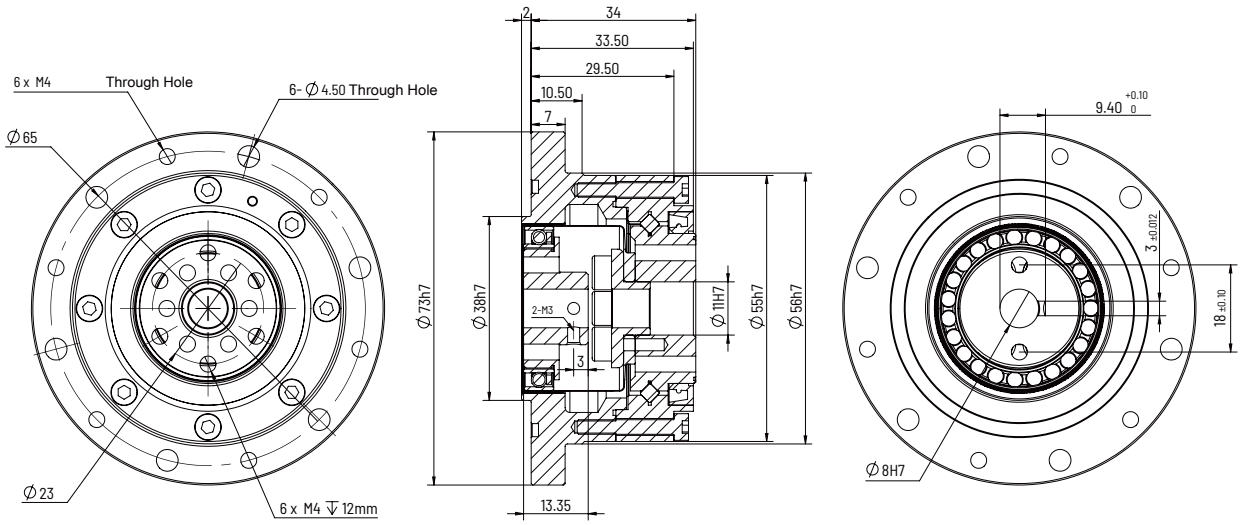
Standard Strain Wave Gear



Diameter: Φ73mm | Thickness: 36mm | Weight≤0.513kg

Drawing

unit: mm



CS-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	4	5.4	7.8	7.8	
Limit for Repeated Peak Torque[Nm]	9	18	23	28	
Limit for Average Torque[Nm]	6.8	6.9	9.5	11	
Limit for Momentary Peak Torque[Nm]	17	35	47	54	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.033	0.033	0.033	0.033	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

CS-17-I

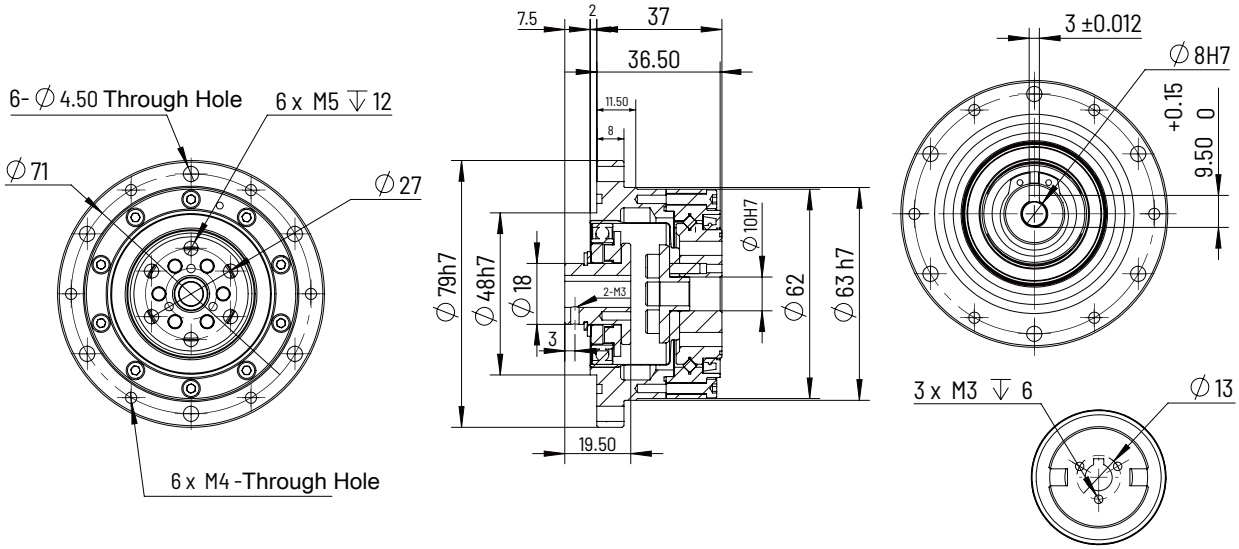
Simplicity Unit Strain Wave Gear



Diameter: Φ79mm | Thickness: 47.44mm | Weight≤0.713kg

Drawing

unit: mm



CS-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	8.8	16	22	24	
Limit for Repeated Peak Torque[Nm]	16	34	43	54	
Limit for Average Torque[Nm]	12	26	27	39	
Limit for Momentary Peak Torque[Nm]	30	70	87	108	
Maximum Input Speed[rpm]	7300	7300	7300	7300	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.079	0.079	0.079	0.079	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

CS-17-II

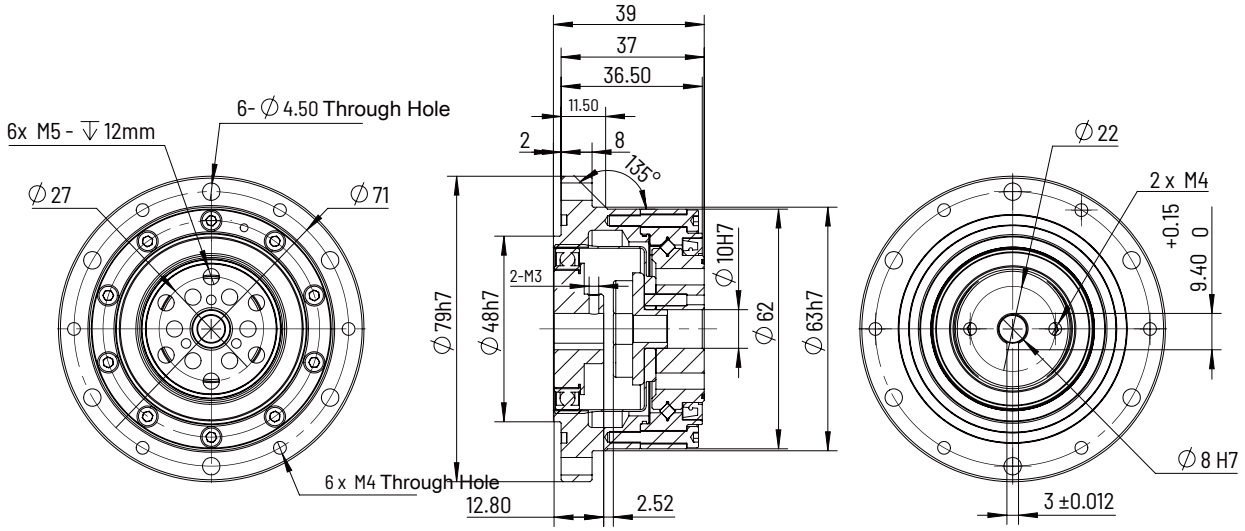
Standard Strain Wave Gear



Diameter: Φ79mm | Thickness: 39mm | Weight≤0.7kg

Drawing

unit: mm



CS-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	8.8	16	22	24	
Limit for Repeated Peak Torque[Nm]	16	34	43	54	
Limit for Average Torque[Nm]	12	26	27	39	
Limit for Momentary Peak Torque[Nm]	30	70	87	108	
Maximum Input Speed[rpm]	7300	7300	7300	7300	
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.079	0.079	0.079	0.079	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

CS-20-I

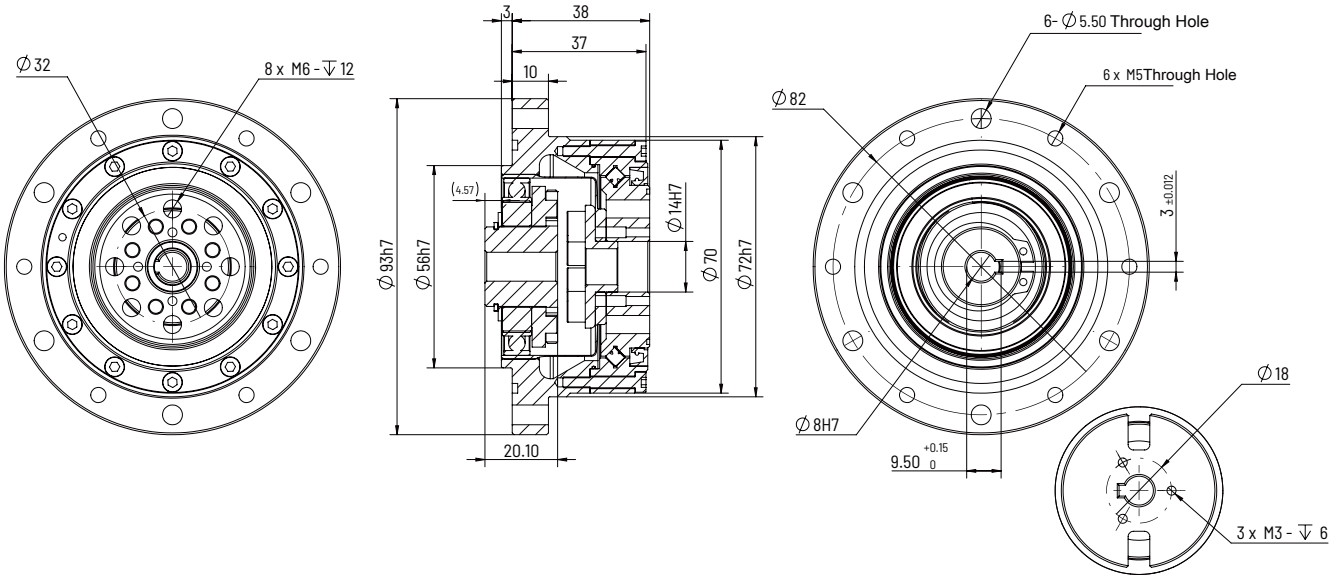


Simplicity Unit Strain Wave Gear

Diameter: Φ93mm | Thickness: 45.5mm | Weight≤1kg

Drawing

unit: mm



CS-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	15	25	34	40	47
Limit for Repeated Peak Torque[Nm]	27	56	74	82	87
Limit for Average Torque[Nm]	20	34	47	49	55
Limit for Momentary Peak Torque[Nm]	50	98	121	147	160
Maximum Input Speed[rpm]	6500	6500	6500	6500	6500
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.193	0.193	0.193	0.193	0.193
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

CS-20-II

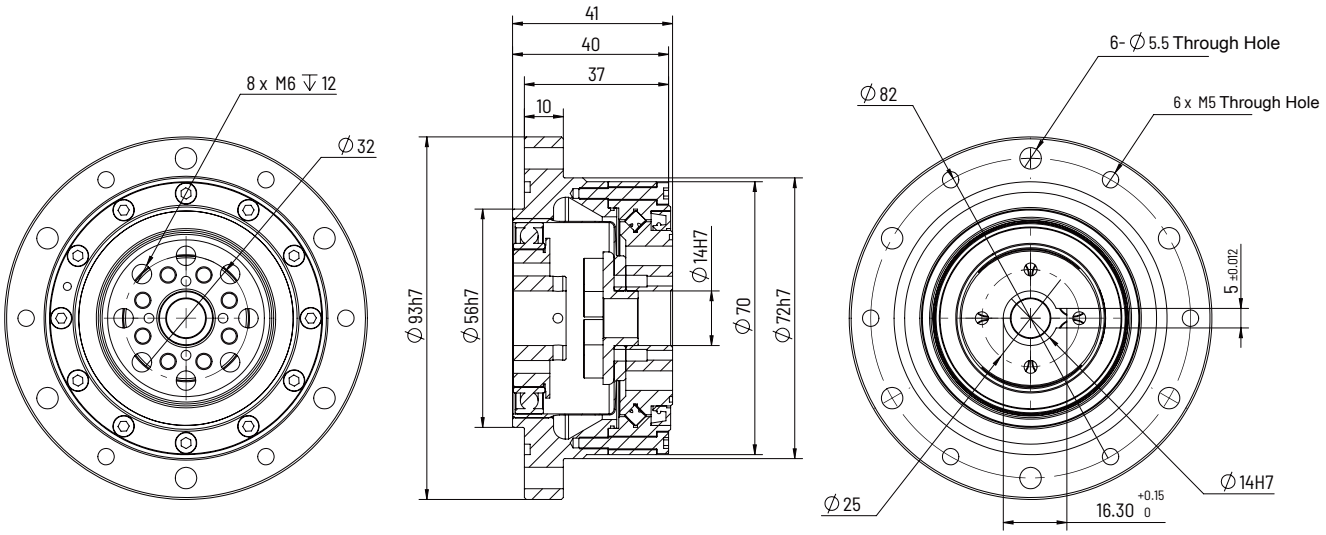
Standard Strain Wave Gear



Diameter: Φ93mm | Thickness: 41mm | Weight≤0.935kg

Drawing

unit: mm



CS-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	15	25	34	40	47
Limit for Repeated Peak Torque[Nm]	27	56	74	82	87
Limit for Average Torque[Nm]	20	34	47	49	55
Limit for Momentary Peak Torque[Nm]	50	98	121	147	160
Maximum Input Speed[rpm]	6500	6500	6500	6500	6500
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.193	0.193	0.193	0.193	0.193
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

CS-25-I

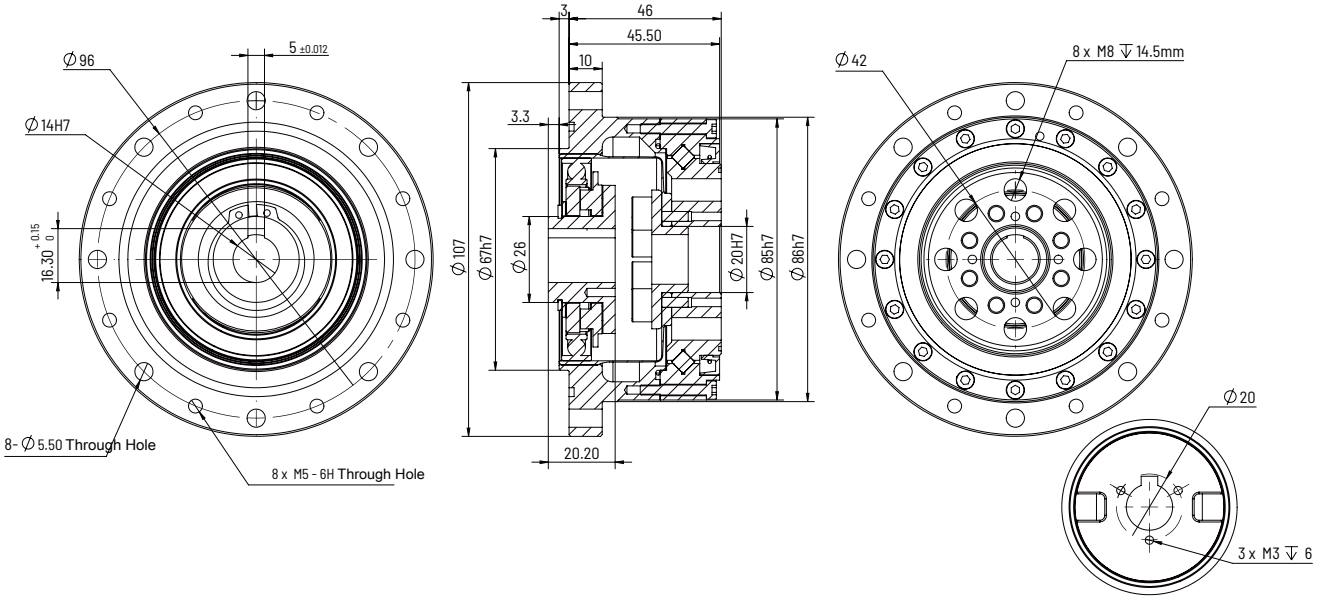
Simplicity Unit Strain Wave Gear



Diameter: Φ107mm | Thickness: 52mm | Weight≤1.503kg

Drawing

unit: mm



CS-25 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	27	39	63	67	78
Limit for Repeated Peak Torque[Nm]	50	98	137	157	167
Limit for Average Torque[Nm]	38	55	87	108	115
Limit for Momentary Peak Torque[Nm]	95	186	255	289	304
Maximum Input Speed[rpm]	5600	5600	5600	5600	5600
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.413	0.413	0.413	0.413	0.413
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

CS-25-II

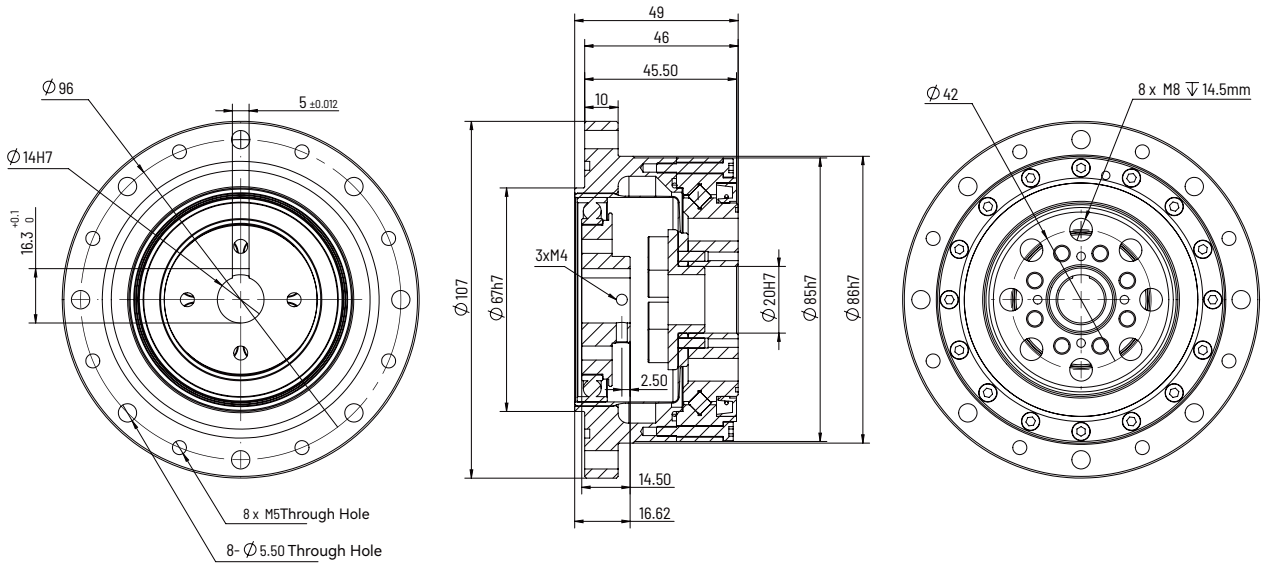
Standard Strain Wave Gear



Diameter: Φ107mm | Thickness: 49mm | Weight≤1.5kg

Drawing

unit: mm



CS-25 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	27	39	63	67	78
Limit for Repeated Peak Torque[Nm]	50	98	137	157	167
Limit for Average Torque[Nm]	38	55	87	108	115
Limit for Momentary Peak Torque[Nm]	95	186	255	289	304
Maximum Input Speed[rpm]	5600	5600	5600	5600	5600
Limit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.413	0.413	0.413	0.413	0.413
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

CS-32-I

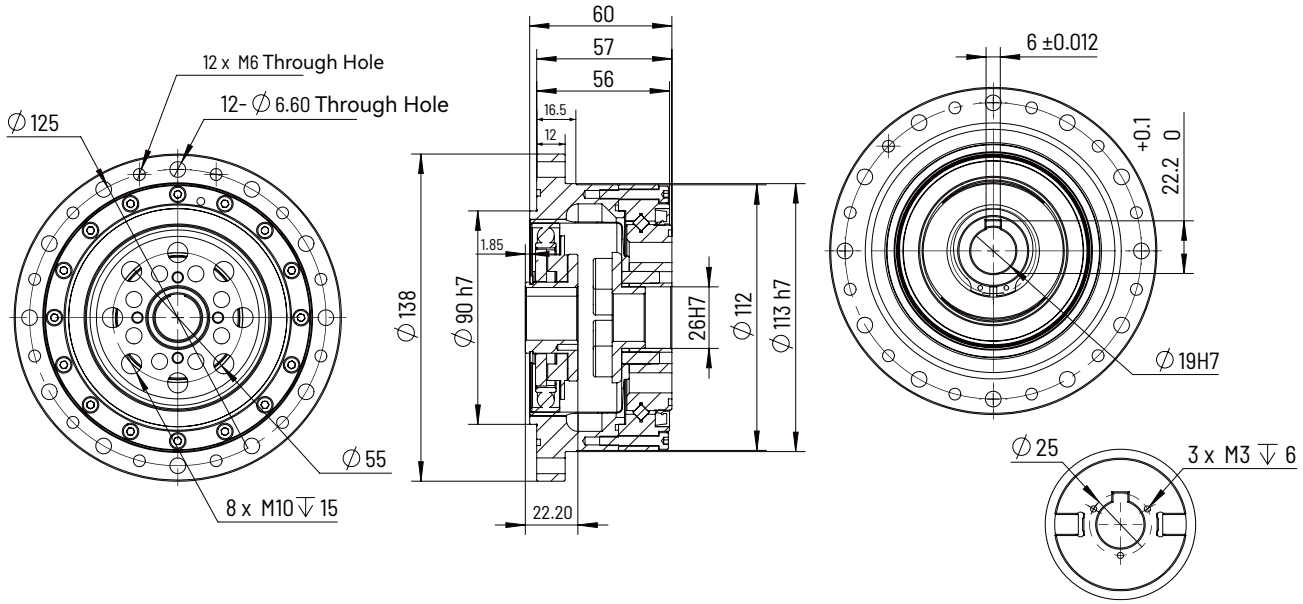


Simplicity Unit Strain Wave Gear

Diameter: Φ138mm | Thickness: 62mm | Weight≤3.095kg

Drawing

unit: mm



CS-32 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		76	118	137	152
Limit for Repeated Peak Torque[Nm]		216	304	353	353
Limit for Average Torque[Nm]		108	167	216	240
Limit for Momentary Peak Torque[Nm]		382	568	647	686
Maximum Input Speed[rpm]		4800	4800	4800	4800
Limit for Average Input Speed[rpm]		3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]		1.69	1.69	1.69	1.69
Backlash[Arcsec]		≤20	≤10	≤10	≤10

STRAIN WAVE DRIVE REDUCERS

CS-32-II

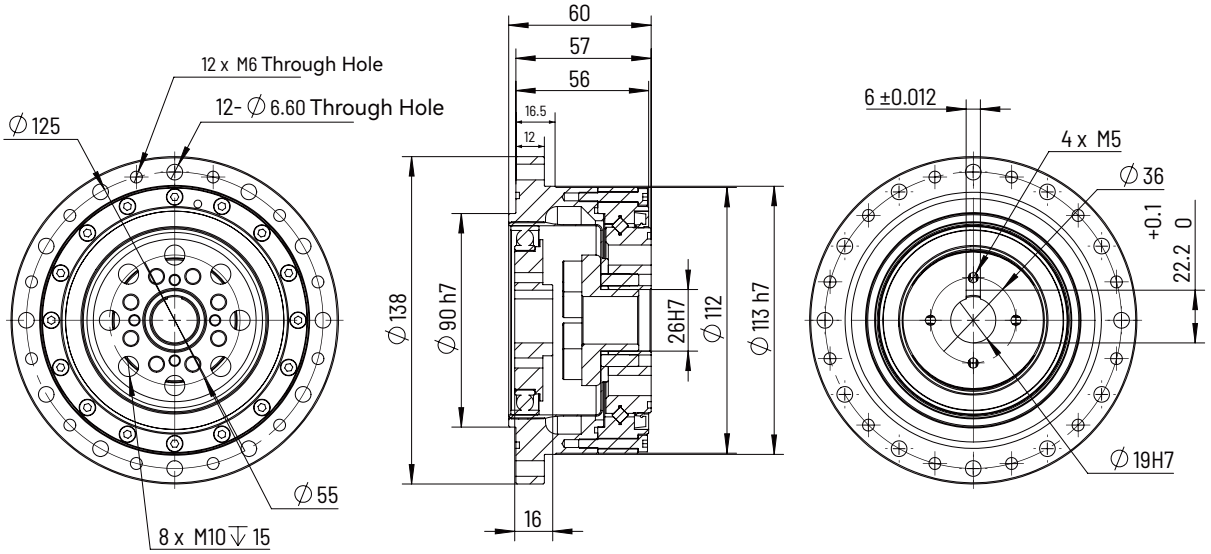
Standard Strain Wave Gear



Diameter: Φ138mm | Thickness: 60mm | Weight≤3.1kg

Drawing

unit: mm



CS-32 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		76	118	137	152
Limit for Repeated Peak Torque[Nm]		216	304	353	353
Limit for Average Torque[Nm]		108	167	216	240
Limit for Momentary Peak Torque[Nm]		382	568	647	686
Maximum Input Speed[rpm]		4800	4800	4800	4800
Limit for Average Input Speed[rpm]		3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]		1.69	1.69	1.69	1.69
Backlash[Arcsec]		≤20	≤10	≤10	≤10

Ultramicroscale Strain Wave Gear



CS Mini-2.5



Strain Wave Gear

Advantages in the Application of Dexterous Hands

The hands of humanoid robots require high flexibility, precise movements, a compact structure, lightweight, low noise, and long lifespan.Considering all factors, the Strain Wave Gear is the most suitable reducer device, with the following outstanding features:

Comparison with the Lead Screw in Dexterous Hands:

Parameter	Strain Wave Gear	Lead Screw	Win
Accuaracy	≤0.005°	0.05mm	Strain Wave Gear
Torque Density	150-200N.m/kg	30-50N.m/kg	Strain Wave Gear
Space	Compact	Requires Additional Structure	Strain Wave Gear
Response Time	10ms	50ms	Strain Wave Gear
Life	500,000 Times	200,000 Times	Strain Wave Gear
Stucture	Simple	Complex	Strain Wave Gear
Cost	In mass production, the Strain Wave Gear is not more expensive than the lead screw, and their cost-pe ormance ratios are essentially comparable.		

STRAIN WAVE DRIVE REDUCERS

CS Mini-2.5

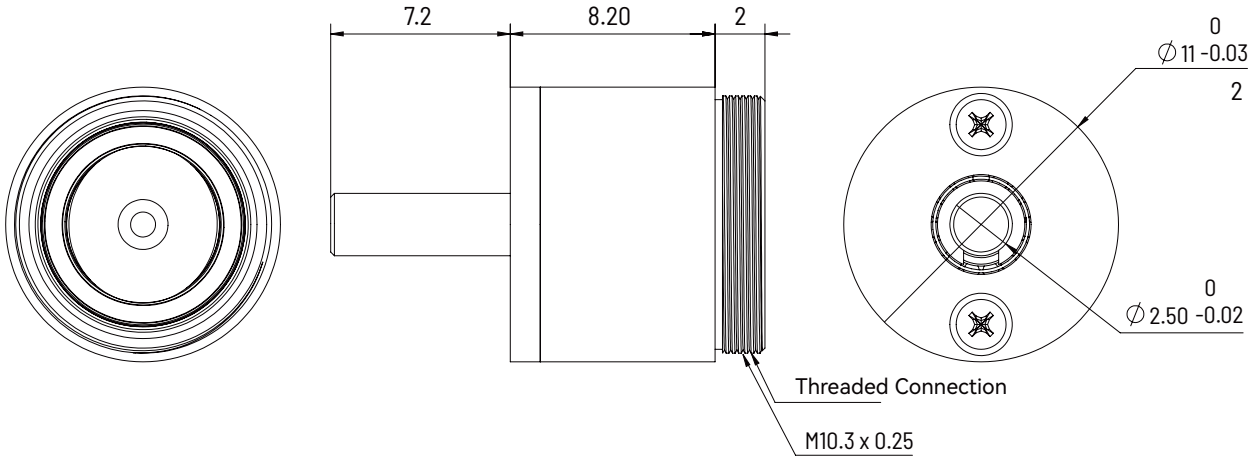
Ultramicroscale Strain Wave Gear



Diameter: 11×11mm | Thickness: 17.4mm | Weight5.9g

Drawing

unit: mm



CS Mini-2.5 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	0.06	0.10			
Limit for Repeated Peak Torque[Nm]	0.13	0.21			
Limit for Average Torque[Nm]	0.10	0.13			
Limit for Momentary Peak Torque[Nm]	0.21	0.40			
Maximum Input Speed[rpm]	7000	7000			
Limit for Average Input Speed[rpm]	3500	3500			
Moment of Inertia [Ix10^-4 kgm^2]					
Backlash[Arcsec]	≤20	≤20			



CS mini
SERIES PRODUCTS
Strain Wave Drive Reducers

STRAIN WAVE DRIVE REDUCERS

CS Mini-03

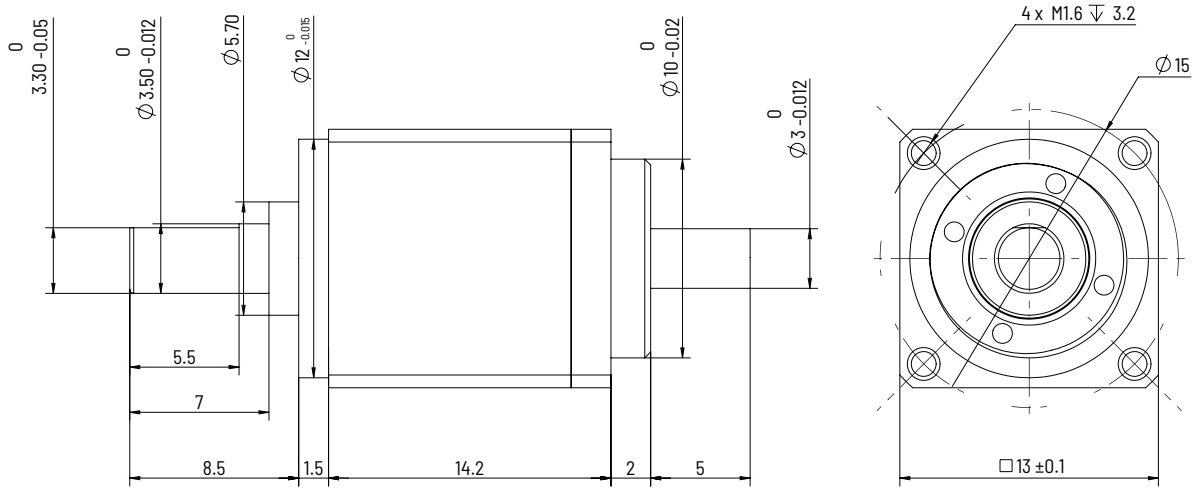
Strain Wave Gear



Diameter: 13×13mm | Thickness: 31.2mm | Weight≤0.017kg

Drawing

unit: mm



CS Mini-03 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	0.06	0.11			
Limit for Repeated Peak Torque[Nm]	0.13	0.21			
Limit for Average Torque[Nm]	0.1	0.13			
Limit for Momentary Peak Torque[Nm]	0.22	0.41			
Maximum Input Speed[rpm]	10000	10000			
Limit for Average Input Speed[rpm]	6500	6500			
Moment of Inertia [Ix10^-4 kgm^2]					
Backlash[Arcsec]	≤20	≤20			

STRAIN WAVE DRIVE REDUCERS

CS Mini-05 Strain Wave Gear

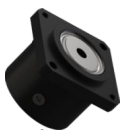
Diameter: ≤22mm | Thickness: ≤22mm | Weight≤0.042kg

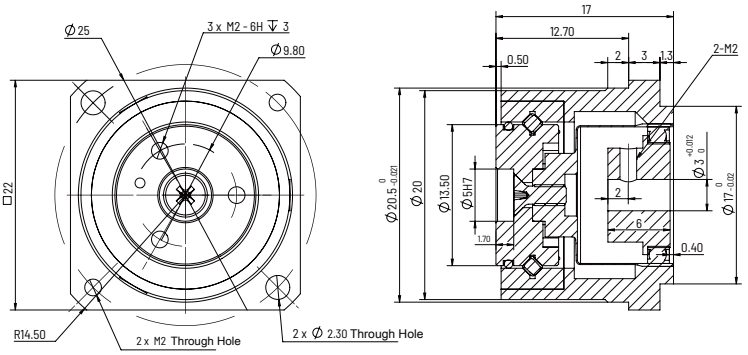
Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	0.25	0.4	0.52		
Limit for Repeated Peak Torque[Nm]	0.5	0.9	1.35		
Limit for Average Torque[Nm]	0.38	0.53	0.84		
Limit for Momentary Peak Torque[Nm]	0.9	1.8	2.5		
Maximum Input Speed[rpm]	10000	10000	10000		
Limit for Average Input Speed[rpm]	6500	6500	6500		
Moment of Inertia [Ix10^-4 kgm^2]	2.5x10^-4	2.5x10^-4	2.5x10^-4		
Backlash[Arcsec]	≤20	≤20	≤20		

Drawing unit: mm

2XH-A

Input Hole & Output Flange

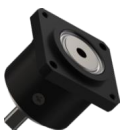


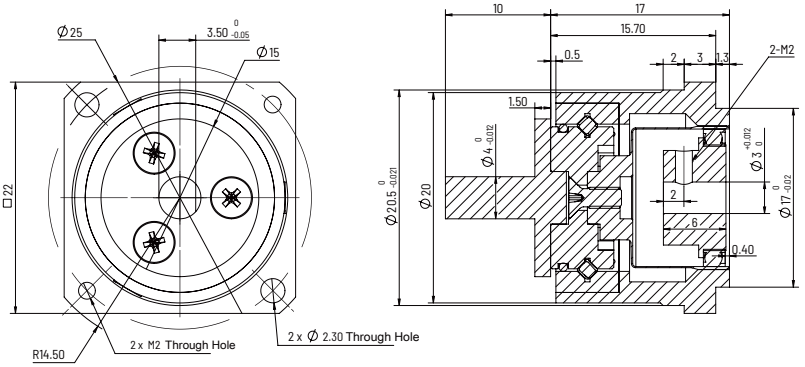


Drawing unit: mm

2XH-B

Input Hole & Output Sharft

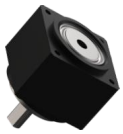


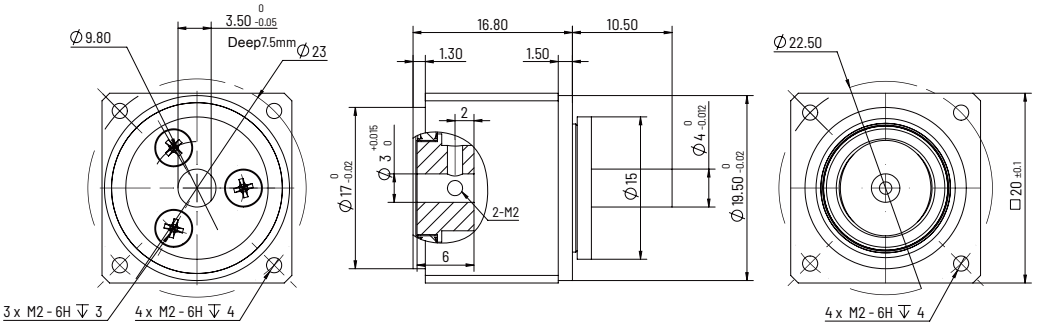


Drawing unit: mm

1U-C

Input Hole & Output Sha t

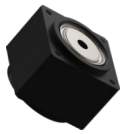


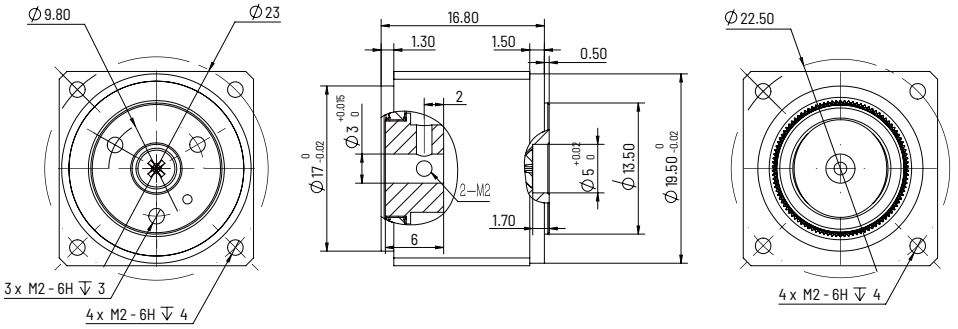


Drawing unit: mm

1U-D

Input Hole & Output Flange



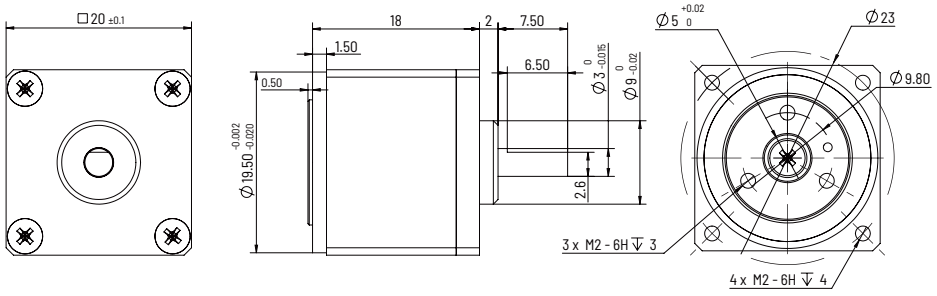


Drawing unit: mm

1U-E

Input Sharft & Output Flange



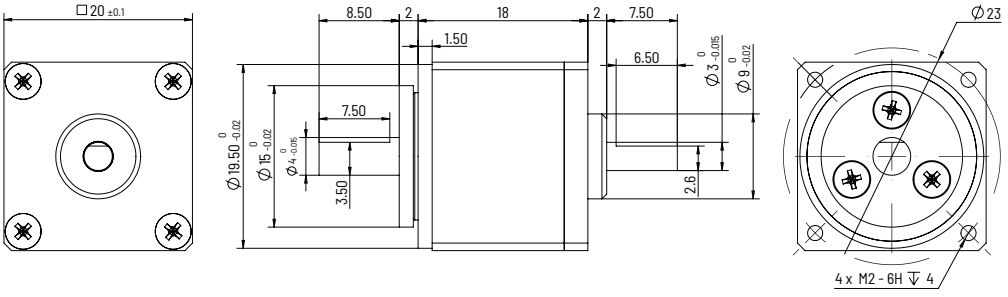


Drawing unit: mm

1U-U

Input Sharft & Output Sharft





STRAIN WAVE DRIVE REDUCERS

CS Mini-08 Strain Wave Gear

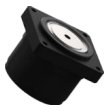
Diameter: ≤50mm | Thickness: 26.5mm | Weight≤0.106kg

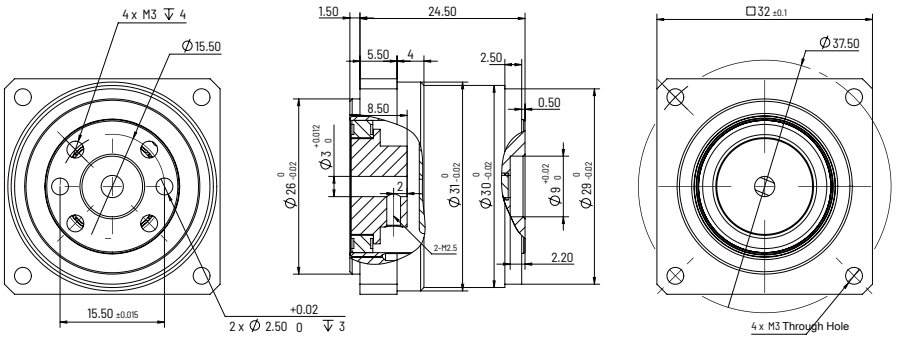
Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	0.9	1.8		2.4	
Limit for Repeated Peak Torque[Nm]	1.8	3.3		4.8	
Limit for Average Torque[Nm]	1.4	2.3		3.3	
Limit for Momentary Peak Torque[Nm]	3.3	6.6		9	
Maximum Input Speed[rpm]	8500	8500		8500	
Limit for Average Input Speed[rpm]	3500	3500		3500	
Moment of Inertia [Ix10^-4 kgm^2]	3.2x10^-3	3.2x10^-3		3.2x10^-3	
Backlash[Arcsec]	≤20	≤20		≤20	

Drawing unit: mm

2XH-A

Input Hole & Output Flange



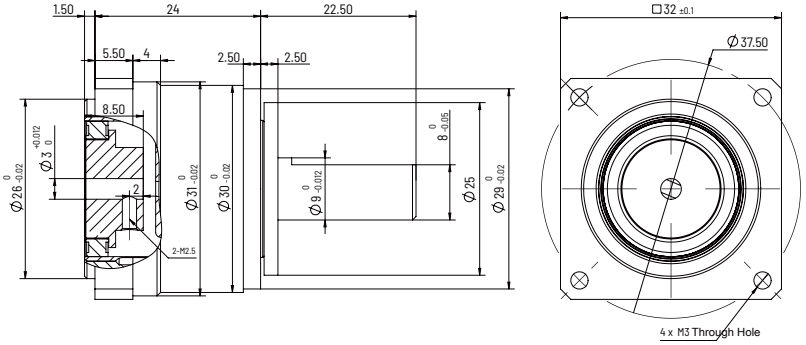


Drawing unit: mm

2XH-B

Input Hole & Output Sharft



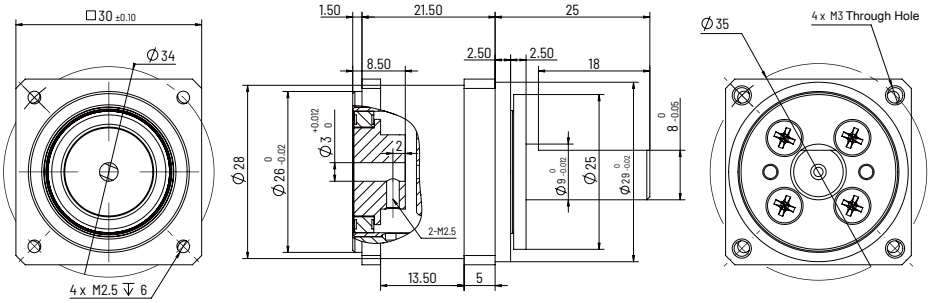


Drawing unit: mm

1U-C

Input Hole & Output Sharft

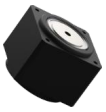


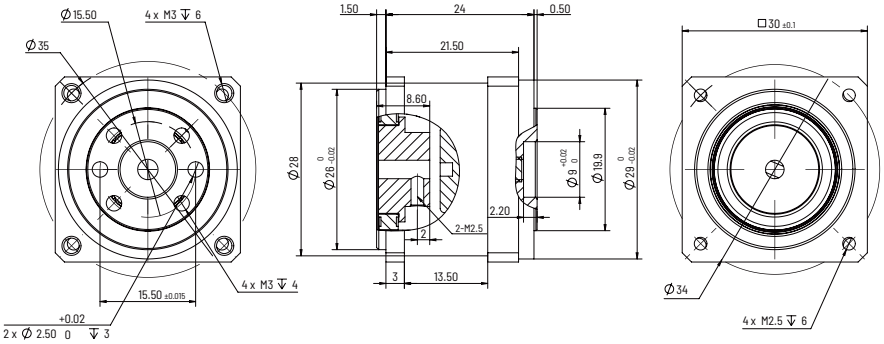


Drawing unit: mm

1U-D

Input Hole & Output Flange



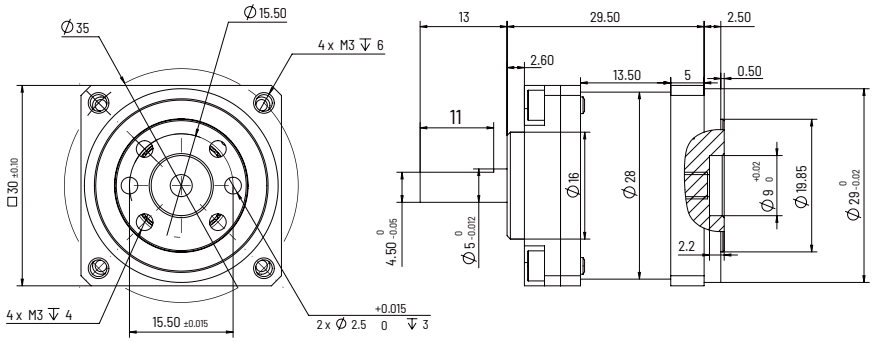


Drawing unit: mm

1U-E

Input Sharft & Output Flange



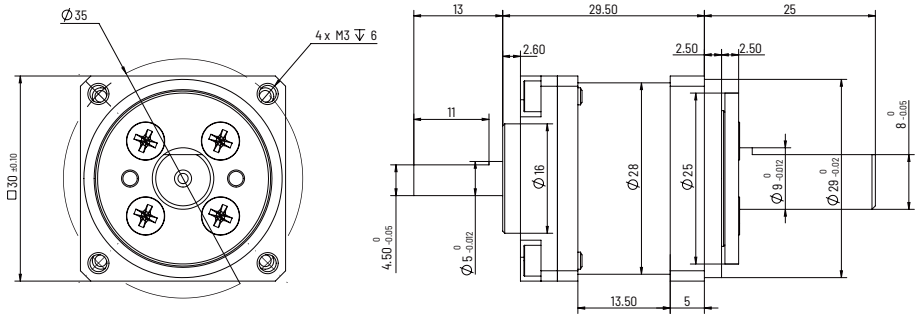


Drawing unit: mm

1U-U

Input Sharft & Output Sharft

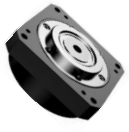


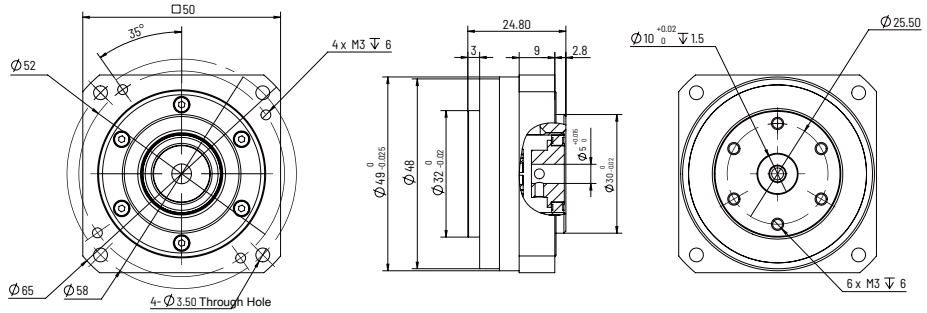


Drawing unit: mm

2UP

Input Hole & Output Flange





STRAIN WAVE DRIVE REDUCERS

CS Mini-11 Strain Wave Gear

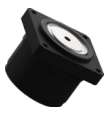
Diameter: ≤43mm | Thickness: ≤40mm | Weight≤0.2kg

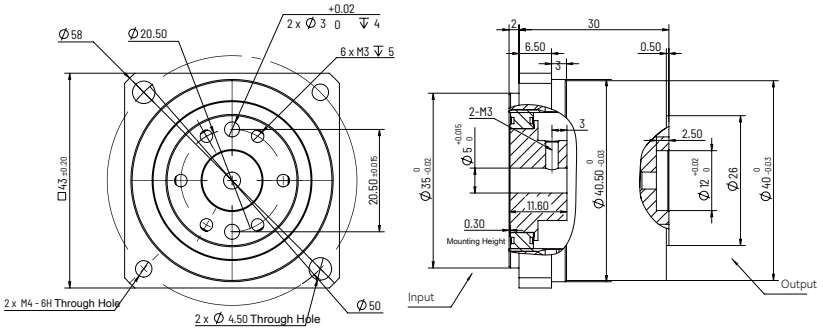
Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	2.2	3.5		5	
Limit for Repeated Peak Torque[Nm]	4.5	8.3		11	
Limit for Average Torque[Nm]	3.4	5.5		8.9	
Limit for Momentary Peak Torque[Nm]	8.5	17		25	
Maximum Input Speed[rpm]	8500	8500		8500	
Limit for Average Input Speed[rpm]	3500	3500		3500	
Moment of Inertia [Ix10^-4 kgm^2]	1.4x10^-2	1.4x10^-2		1.4x10^-2	
Backlash[Arcsec]	≤20	≤20		≤20	

Drawing unit: mm

2XH-A

Input Hole & Output Flange



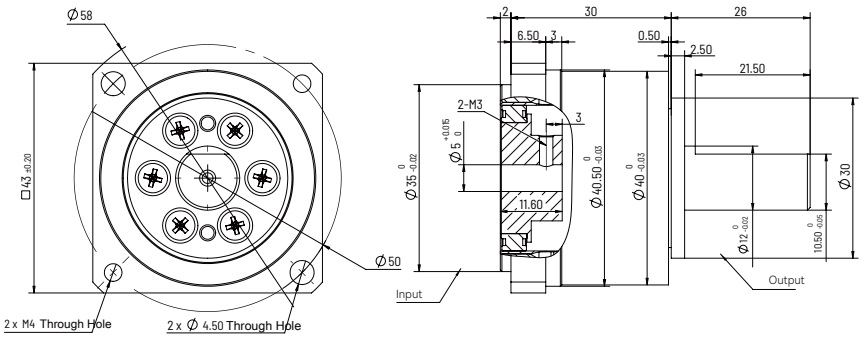


Drawing unit: mm

2XH-B

Input Hole & Output Sharft



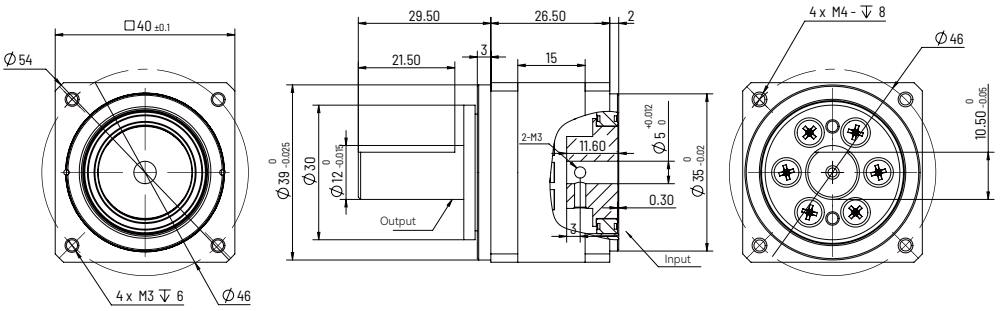


Drawing unit: mm

1U-C

Input Hole & Output Sharft

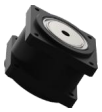


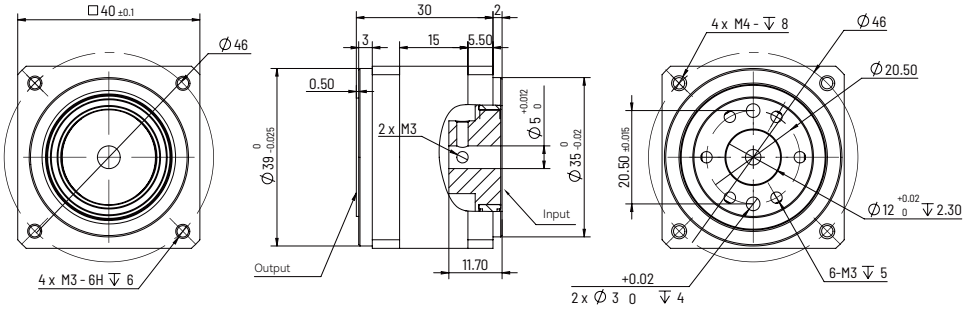


Drawing unit: mm

1U-D

Input Hole & Output Flange



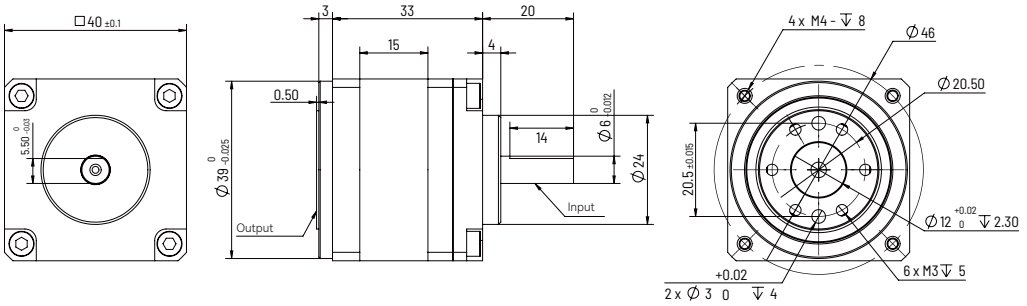


Drawing unit: mm

1U-E

Input Sharft & Output Flange



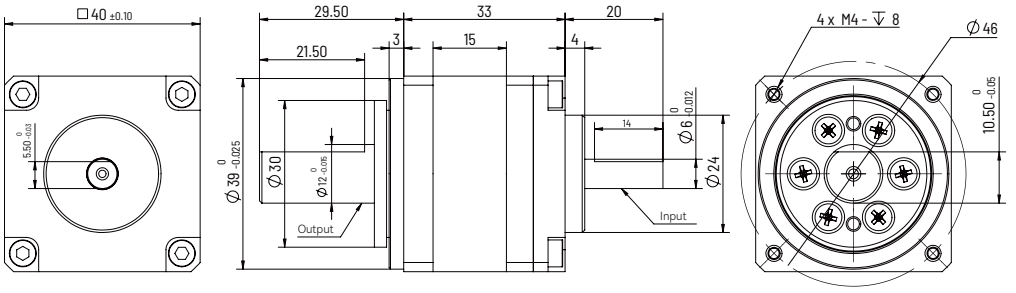


Drawing unit: mm

1U-U

Input Sharft & Output Sharft





CD

STRAIN WAVE DRIVE REDUCERS

CD Series

Model	Gear Ratio	Page
CD-14	14: 30/50/80/100	83
CD-17	17: 50/80/100	84
CD-20	20: 30/50/80/100/120	85



Features and Applications

The CD series meets market demands by inheriting the excellent pe ormance of traditional products while achieving bold shape designs that emphasize the lightweight and compact characteristics of harmonic gear drives.

Features

- Compact
- High Rigidity
- High Torque Capacity



CD SERIES PRODUCTS

Strain Wave Drive Reducers

Product Model Description

CD-20-100

Series	Model	Gear Ratio						-	-
CD	14	30	50	80	100			-	-
	17		50	80	100				
	20	30	50	80	100	120			

Series	Model	Ratio	Rate Torque at 2000 rpm [N.m]	Limit for Repeated Peak Torque [N.m]	Limit for Average Torque [N.m]	Limit for Momentary Peak Torque [N.m]	Maximum Input Speed [r/min]	Limit for Average Input Speed [r/min]	Moment of Inertia (1/4GD^2) [kgcm^2]	Backlash [Arc sec]	Life [hour]
CD	CD-14	30	2.2	10.2	2.88	14.4	8500	3500	0.021	≤20	7000
		50	3.7	12	4.8	24					
		80	5.4	16	7.7	31				≤10	10000
		100	5.4	19	7.7	31					
	CD-17	50	11	23	18	48	7300	3500	0.054	≤20	7000
		80	15	29	19	55				≤10	10000
		100	16	37	27	55					
	CD-20	30	10.2	23.4	14.4	41.4	6500	3500	0.09	≤20	7000
		50	17	39	24	69					
		80	24	51	33	76				≤10	10000
		100	28	57	34	76					
		120	28	61	34	95					

STRAIN WAVE DRIVE REDUCERS

CD-14

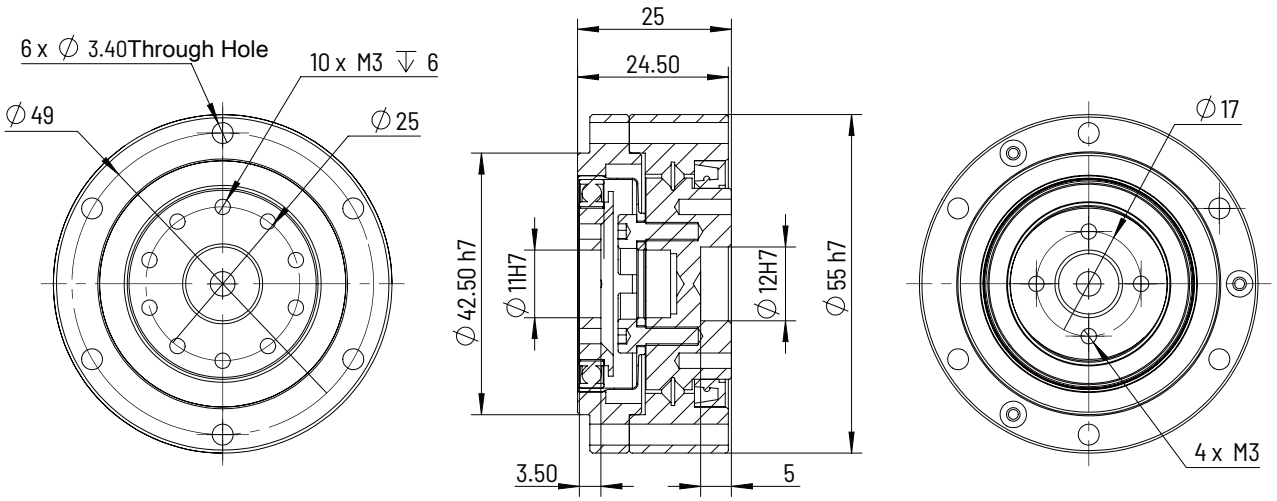
Strain Wave Gear



Diameter: Φ55mm | Thickness: 25mm | Weight≤0.35kg

Drawing

unit: mm



CD-14 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	2.2	3.7	5.4	5.4	
Limit for Repeated Peak Torque[Nm]	10.2	12	16	19	
Limit for Average Torque[Nm]	2.88	4.8	7.7	7.7	
Limit for Momenta. Peak Torque[Nm]	14.4	24	31	31	
Maximum Input Speed[rpm]	8500	8500	8500	8500	
imit for Average Input Speed[rpm]	3500	3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]	0.021	0.021	0.021	0.021	
Backlash[Arcsec]	≤20	≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

CD-17

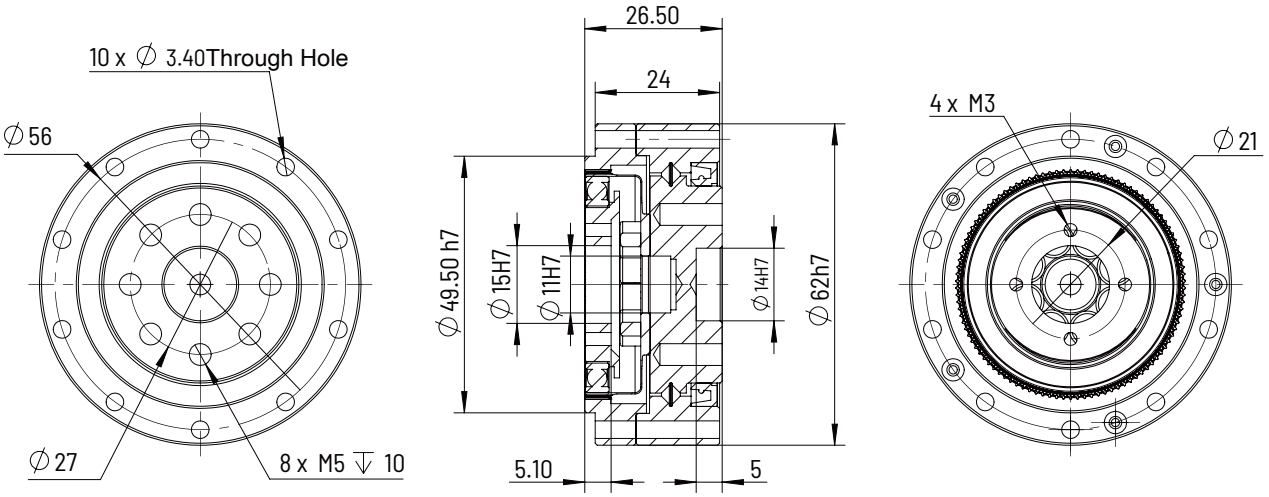
Strain Wave Gear



Diameter: Φ62mm | Thickness: 26.5mm | Weight≤0.421kg

Drawing

unit: mm



CD-17 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]		11	15	16	
Limit for Repeated Peak Torque[Nm]		23	29	37	
Limit for Average Torque[Nm]		18	19	27	
Limit for Momentary Peak Torque[Nm]		48	55	55	
Maximum Input Speed[rpm]		7300	7300	7300	
imit for Average Input Speed[rpm]		3500	3500	3500	
Moment of Inertia [Ix10^-4 kgm^2]		0.054	0.054	0.054	
Backlash[Arcsec]		≤20	≤10	≤10	

STRAIN WAVE DRIVE REDUCERS

CD-20

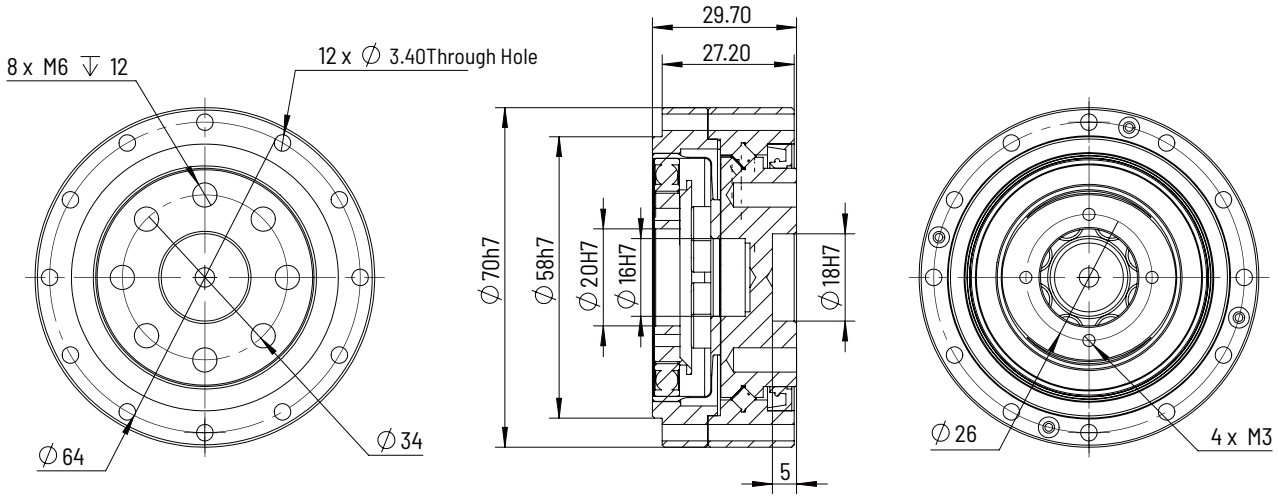
Strain Wave Gear



Diameter: Φ70mm | Thickness: 29.7mm | Weight≤0.591kg

Drawing

unit: mm



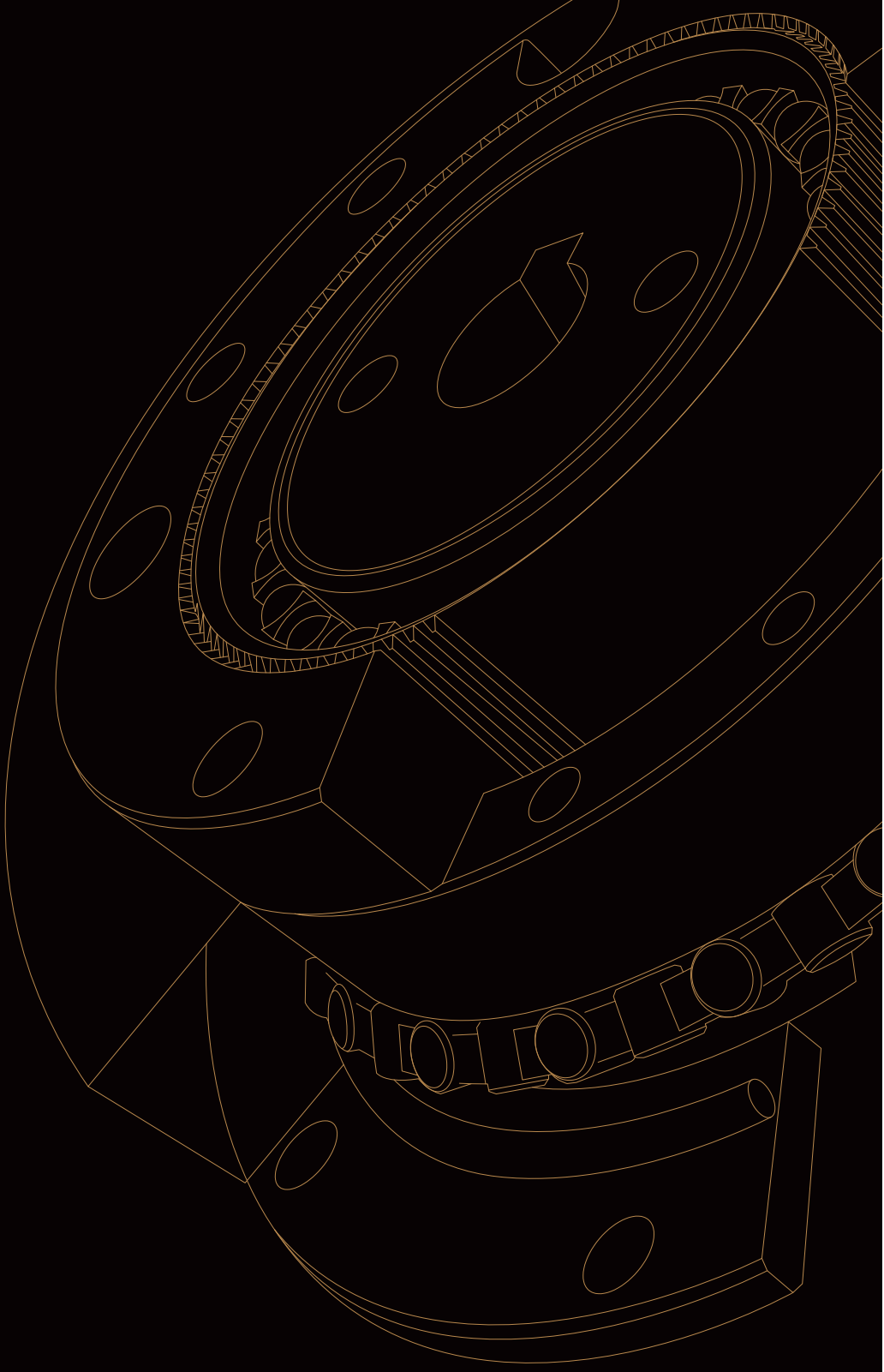
CD-20 [Parameter]

Gear Ratio	30	50	80	100	120
Rate Torque at 2000 rpm [Nm]	10.2	17	24	28	28
Limit for Repeated Peak Torque[Nm]	23.4	39	51	57	61
Limit for Average Torque[Nm]	14.4	24	33	34	34
Limit for Momentary Peak Torque[Nm]	41.4	69	76	76	95
Maximum Input Speed[rpm]	6500	6500	6500	6500	6500
imit for Average Input Speed[rpm]	3500	3500	3500	3500	3500
Moment of Inertia [Ix10^-4 kgm^2]	0.09	0.09	0.09	0.09	0.09
Backlash[Arcsec]	≤20	≤20	≤10	≤10	≤10



SITO

SITO Drive Technology Co.,Ltd.



鑫拓智能

STRAIN WAVE DRIVE REDUCERS PRODUCT SELECTION MANUAL

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